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The electric vehicle routing problem with time windows, partial recharges, and covering locations

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Abstract

This research studies the electric vehicle routing problem with time windows, partial recharges, and covering locations (EVRPTW-PR-CL), as an extension of the electric vehicle routing problem with time windows and partial recharges (EVRPTW-PR), where covering locations (CLs) are facilities equipped with parcel lockers (PLs) and charging stations (CSs). The presence of PLs offers customers an alternative delivery option, where they are provided incentives to collect their parcels themselves, called self-pickup (SP) services. The objective is to seek routing plans that minimize the sum of travel costs, fixed operational costs for used EVs and CLs, and compensation paid to customers served by SP. To solve the problem, we derive a mixed-integer programming model and design an effective variable neighborhood search (VNS) algorithm coupled with problem-specific neighborhood operators, a dynamic programming procedure for optimal CS insertions, and a tailored set partitioning formulation (SPF) to enhance solution quality by utilizing collected routes so far. Numerical experiments are conducted on benchmark instances. VNS not only provides new best-known EVRPTW-PR solutions but also solves EVRPTW-PR-CL instances efficiently. Lastly, we present the effects of delivery options and compensation, offering insights that help decision makers design more sustainable and cost-effective last-mile delivery networks.

Keywords: electric vehicle routing problem; variable neighborhood search; covering location; delivery option; compensation rate

1. Introduction

Consumers have become increasingly familiar with online shopping habits, resulting in the rapid growth of the e-commerce industry (Sadati et al., 2022). Over 26 million e-commerce sites were launched in 2018, serving more than 2.14 billion users who account for approximately 27.6% of the world's population (Kin, 2022). Furthermore, e-commerce sales are expected to surpass US$ 6.388

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trillion by 2024 for an annual growth rate of 13.5% (Statista, 2022). This boom in e-commerce has significantly posed economic and environmental concerns for developing sustainable last-mile delivery networks. Last-mile delivery activities predominantly rely on fossil-fuel vehicles recognized for being both costly and environmentally detrimental (Cantillo et al., 2022). These activities contribute significantly to total logistics costs, accounting for up to 75% (Gevaers et al., 2009) and contributing to about 7% of total global greenhouse gas emissions (UNCTAD, 2017). Alternative vehicles, such as electric vehicles (EVs), are utilized to reduce emissions, while alternative delivery methods such as self-pickup (SP) services are offered to cut operation costs (Boysen et al., 2021).

Addressing environmental concerns involves a shift from vehicles powered by fossil fuels to alternative vehicles such as EVs. The adoption of EVs, recognized for their zero carbon emission, mitigates the negative environmental impacts in last-mile delivery (de Mello Bandeira et al., 2019). Logistics providers such as DHL, Amazon, and FedEx have recently launched EV fleets in their last-mile delivery networks. The adoption of EVs was first studied by Conrad and Figliozzi (2011) to provide a more sustainable routing solution. Later papers further extended the concepts of the electric vehicle routing problem (EVRP). In particular, the limited driving range of EVs is addressed by allowing for the recharging of vehicle batteries at charging stations (CSs) (Schneider et al., 2014). Partial recharge helps mitigate the recharging times and battery costs (Keskin and Çatay, 2016).

Attended home delivery (AHD) is a common delivery method, where customers receive parcels at their private locations (Sadati et al., 2022). AHD faces challenges in operational costs and time efficiency. Customer absence and traffic congestion further contribute to delivery failures, leading to increased air pollution, higher logistics costs, and decreased customer satisfaction levels. Logistics companies address such challenges by exploring alternative delivery methods (Boysen et al., 2021). Notably, over 20 developed countries have successfully integrated parcel lockers (PLs) into their last-mile delivery networks (Deutsch and Golany, 2018). This has led to the rise of the SPU service, where customers collect parcels from designated PLs (Enthoven et al., 2020). Furthermore, incentives should be offered to address customer dissatisfaction with the SPU service (Dietl et al., 2024).

We then introduce a new variant of the EVRP with time windows and partial recharges (EVRPTW-PR) proposed by Keskin and Çatay (2016), which incorporates PLs used for SP services and CSs for recharging EV batteries. In practice, EVs often recharge their batteries while serving demand points (Yavuz and Çapar, 2017; Koyuncu and Yavuz, 2019), with PLs typically positioned near CSs for EVs in order to facilitate efficient recharging during parcel deliveries (Vukićević et al., 2023). Consequently, this paper presents covering locations (CLs) as facilities equipped with both PLs and CSs. Finally, compensation for customers opting for the SP service is included as a cost component in the objective function, and the distance from customer locations to their assigned PLs is kept within a reasonable range to ensure customer accessibility. Our main contributions are highlighted as follows.

- We present the electric vehicle routing problem with time windows, partial recharges, and covering locations (EVRPTW-PR-CL), incorporating the use of EVs and CLs with two delivery options. The objective is to find optimal routing plans that minimize travel costs, fixed costs for used EVs, fixed operational costs for CLs used for serving customers, and especially compensation costs.

- We formulate a mixed-integer linear programming (MILP) model to solve small EVRPTW-PR-CL instances. We also design an effective variable neighborhood search (VNS), incorporating problem-specific neighborhood operators and a tailored set partitioning formulation (SPF) to solve larger instances.
- We test VNS on solving both EVRPTW-PR benchmark instances and EVRPTW-PR-CL instances. The results show that VNS provides new best-known solutions (BKSs) for EVRPTW-PR, while the algorithm effectively solves EVRPTW-PR-CL within a reasonable computational time.
- We finally conduct sensitivity analyses related to the adoption of CLs equipped with PLs and CSs with different settings of customer distribution, compensation rate, and covering level. These results provide managerial insights for decision makers in their networks.

The remaining parts are structured as follows. Section 2 reviews papers related to EVRP variants and delivery options. Section 3 provides a formal description of the EVRPTW-PR-CL network along with a mathematical model to solve EVRPTW-PR-CL. Section 4 proposes a VNS algorithm that incorporates operators designed for specifically addressing this network. Section 5 conducts numerical experiments to evaluate the performance of VNS and analyzes the impacts of delivery options. Section 6 gives managerial insights by considering specific characteristics with different settings. Section 7 offers conclusions and future research direction.

2. Literature review

This section briefly reviews related works in the literature to address the gaps between existing research and our proposed problem. Section 2.1 describes EVRP variants in other works. Section 2.2 discusses delivery options in last-mile delivery, emphasizing the presence of PLs and their location decisions.

2.1. Electric vehicle routing problems

The use of EVs in the vehicle routing problem (VRP) was first introduced by Conrad and Figliozzi (2011), where EVs are allowed to recharge up to a minimum of 80% capacity with a constant recharging time, formulated by a mixed-integer nonlinear model. Schneider et al. (2014) formalized EVRP with time windows (EVRPTW), in which a set of customers must be served within their predefined time windows. EVs are always recharged fully whenever they visit a recharging station, called the full recharging strategy. EVRPTW instances were newly generated and then solved by an MILP model. Keskin and Çatay (2016) suggested the partial recharging strategy, which allows EVs to recharge their batteries up to any level. Desaulniers et al. (2016) developed a branch and price and cut algorithm to generate feasible routes to solve EVRPTW variants. Erdelić and Carić (2022) then proposed an adaptive large neighborhood search algorithm to obtain high-quality solutions. EVRP variants focus on modeling the recharging process realistically and integrating EVs into new logistics operations.

Goeke and Schneider (2015) utilized both EVs and internal combustion vehicles in VRP with time windows (VRPTW), introducing a more realistic formulation for consumption rates that improves routing plans. Hiermann et al. (2019) extended the former problem by adding plug-in hybrid vehicles for reduced operational costs. Cortés-Murcia et al. (2019) proposed a new strategy, employing alternative means during EV recharging (e.g., walking, bikes, and drones). Nonlinear approximations for comprehensive recharging consumption were derived by Montoya et al. (2017) and Rastani and Çatay (2023). Keskin and Çatay (2018) discussed recharging technologies with different speeds and costs, while Keskin et al. (2019) considered waiting times at public CSs. Subsequently, Froger et al. (2022) expanded this problem by accounting for the capacity for multiple EVs to visit a CS simultaneously and the synchronization of recharging schedules. The use of PLs was also investigated in EVRPTW (Yu et al., 2023b; Vukićević et al., 2023). For a comprehensive overview, readers can refer to Erdelić and Carić (2019).

2.2. Delivery options in last-mile delivery

To alleviate the burden in last-mile delivery operations, Boysen et al. (2021) conducted a review of alternative delivery options and additional customer requirements spanning past, present, and future last-mile concepts. Some highlighted studies are summarized as follows. Reyes et al. (2017) presented the concept of roaming delivery locations within the framework of VRPTW, involving multiple potential locations for serving each customer. Ozbaygin et al. (2017) further utilized customer car trunks as the roaming delivery locations, which was solved by a branch-and-price algorithm. A dynamic version of the former problem was studied by Ozbaygin and Savelsbergh (2019), addressing changes in customer routines during the delivery process. The dynamic version was solved using the algorithm of Ozbaygin et al. (2017) with an iterative reoptimization heuristic.

Zhou et al. (2018) presented two delivery options: home delivery and customer pickup within the framework of two-echelon VRP. For customer pickup, packages are collected at shared locations. Deutsch and Golany (2018) implemented PLs in a last-mile delivery network, reducing overall logistics costs by 66% compared to traditional methods. Enthoven et al. (2020) introduced CLs equipped with PLs, offering SP services, where customers can receive compensation for picking up packages at CLs within a predefined radius. Vukićević et al. (2023) also looked at CLs in a last-mile delivery network with an EV, which minimizes the total delivery time by using only one EV. Dumez et al. (2021) proposed an extensive network, including private addresses, PLs, and car trunks as delivery locations with varied preference levels. With the presence of PLs, an SP service is provided to serve customers, which is more flexible. Yu et al. (2023b) utilized two delivery options (i.e., the AHD option and the SP service) in the context of EVRPTW. With two delivery options, three types of customers are considered in the problem. Other lockers' properties such as automated PL system (Orenstein et al., 2019) and heterogeneous locker boxes (Grabenschweiger et al., 2021) were also addressed.

Due to decisions needed to be made for PL locations, researchers are also interested in designing last-mile delivery networks. Deutsch and Golany (2018) first designed a last-mile delivery network under the presence of PLs, which optimizes the number, locations, and size of PLs while maximizing total profit; that is, the revenue from customers minus the fixed and operational costs. Luo et al. (2022) then developed another model that minimizes total cost and maximizes customer accessibility by selecting PL locations and assigning customers to specific PLs. Kahr (2022) incorporated stochastic demand and locker availability when determining PL locations and configurations, aiming to maximize the expected number of customers served. Mancini et al. (2023) addressed locker capacity uncertainty in a real-world delivery network in Turin, Italy, using a two-stage stochastic optimization approach. Zhou et al. (2023) developed a decision model to optimize PL locations, balancing operations costs with customer incentives.

Many studies have integrated PL location and routing decisions in last-mile delivery problems, commonly known as location routing problems (LRPs) (Ozyavas et al., 2024). Mara et al. (2021) provided a review of LRP variants. Here, we focus on LRPs involving PLs. Stenger et al. (2012), Zhou et al. (2016), and Veenstra et al. (2018) modeled PLs as customer delivery points, which offer alternative delivery options for serving customers. These models focus on optimizing locker location and routing decisions to distribute parcels to final customers.

Hong et al. (2019) introduced an on-demand delivery system using PLs exclusively to serve customers, formulating a traveling salesman problem model with PL location decisions to optimize travel distance and customer preferences (i.e., penalties for late deliveries and pickup distances). Grabenschweiger et al. (2022) proposed the concept of automatic PLs in multiperiod last-mile delivery operations, first determining opened PLs and then planning routes for each period. Schwerdfeger and Boysen (2022) studied a delivery system where a vehicle is equipped with modular PLs, and PLs are then positioned at suitable parking spots to minimize customer inconvenience. Bonomi et al. (2022) discussed how customer behavior and delivery operations impact environmental outcomes, emphasizing reduced emissions and promoting eco-friendly delivery options with PLs.

The aforementioned studies provide valuable insights into integrating EVs and PLs in delivery operations. While a few works have incorporated both EV adoption and the presence of PLs in last-mile delivery networks (Yu et al., 2023b; Vukićević et al., 2023), they overlooked customer behaviors in delivery services, particularly incentives to promote SP service adoption and customer discomforts. Our study fills these gaps by introducing compensation for customers opting for the SP service and incorporating fixed operational costs for used PLs. We also assume that customers prefer PLs within reasonable distances so as to ensure accessibility. Additionally, our proposed problem involves customer responsibility to contribute to sustainability in delivery activities (Jaller and Pahwa, 2020), resulting in three customer types with two delivery options that align with sustainable practices.

3. Problem description and mathematical model

This problem presents a last-mile delivery network with CLs served by a fleet of EVs through two delivery methods. In particular, Option 1 is the AHD method, in which customers receive parcels directly at their private locations within predefined time windows. Option 2 is the SP service, in which customers collect parcels at one of the designated CLs. For the latter option, incentives can influence customer behavior when choosing delivery options (Dietl et al., 2024). Thus, we assume that some customers are open to being served through Option 2 with compensation to offset any inconvenience from forgoing home delivery. In this model, the compensation is also referred to as the connection cost. Moreover, customers are assigned to a CL based on model decisions, ensuring that the CL is selected within an acceptable distance for customer accessibility, referred to as the coverage radius. The effort required for customers to reach any CL within this radius is then not significantly different. This setting allows solution flexibility without significantly compromising customer satisfaction.

Based on the two delivery options, we propose three types of customers as follows.

- Type 1 customers are served by Option 1 and prioritize traditional and convenience delivery.
- Type 2 customers are served by Option 2 and prefer to collect their parcels at desired PLs.
- Type 3 customers are indifferent to delivery options; that is, served by either Option 1 or Option 2. Indeed, last-mile delivery sustainability is enhanced by encouraging customers to share responsibility for delivery practices (Jaller and Pahwa, 2020), supporting the inclusion of some as Type 3 customers.

While only large companies with extensive resources can provide same-day delivery, many delivery networks still schedule next-day or some-day delivery to serve customers efficiently (Jaller and Pahwa, 2020). In our problem, we assume that all customer requests are confirmed at least one day in advance. Therefore, decisions on delivery options (for Type 3 customers) and the assigned CLs for pickup (for customers served by Option 2) are communicated to customers before the delivery day.

This proposed network is formally defined by a directed graph $G = (N, A)$, where N denotes the set of all nodes and A denotes the set of all arcs. Set $N = N_D \cup N_C \cup N_R$ includes the three subsets as follows.

- $N_D = \{0, 0'\}$ represents the initial and final depots, respectively. We assume that all customer parcels are initially available at the depot, from which EVs distribute them either directly to customers or to associated PLs.
- $N_C = N_H \cup N_S \cup N_F$ represents the set of $n_C = n_H + n_S + n_F$ customers. Here, $N_H = \{1, \dots, n_H\}$ corresponds to n_H Type 1 customers, $N_S = \{n_H + 1, \dots, n_H + n_S\}$ is n_S Type 2 customers, and $N_F = \{n_H + n_S + 1, \dots, n_H + n_S + n_F\}$ includes n_F Type 3 customers
- $N_L = \{1, \dots, n_L\}$ represents the set of n_L CLs. Assume that each CL is equipped with both PLs and CSs, enabling customers to pick up their parcels and EVs to recharge their batteries. In reality, it is quite feasible to locate a PL in proximity to a public CS (Vukićević et al., 2023), allowing EVs to recharge while unloading parcels (Yavuz and Çapar, 2017; Koyuncu and Yavuz, 2019). Each CL possesses a locker capacity Q_L, which can be visited multiple times by EVs. Without loss of generality, the locations of PLs and CSs can be considered independently. In particular, at CLs equipped solely with CSs, we set $Q_L = 0$, indicating no PL capacity. Conversely, for CLs equipped solely with PLs, the charging rate is set to $g = 0$, signifying the absence of CS at these locations.
- For the sake of formulation, we replicate CLs, resulting in a set of replicated CLs (denoted by N_R). Furthermore, the subset of replicated CLs of each CL $i \in N_L$ is denoted as $\mathcal{K}_i \subset N_R$, where $\bigcup_{i \in N_L} \mathcal{K}_i = N_R$. Following Keskin and Çatay (2016), we assume infinite availability of chargers at each CL. We note that the number of visits to each CL can be restricted by limiting the number of replications without loss of generality.

We also denote subsets of N such as $N_1 = N_C \cup N_R$, $N_2 = N_S \cup N_F$, $N_3 = \{0\} \cup N_R$, $N^+ = N \setminus \{0'\}$, and $N^- = N \setminus \{0\}$. Therefore, we define $A = \{(i, j) | i \in N^+, j \in N^-, i \neq j\}$, where each arc $(i, j) \in A$ corresponds with travel cost c_{ij} and travel time t_{ij}. In addition, each node $i \in N$ can be

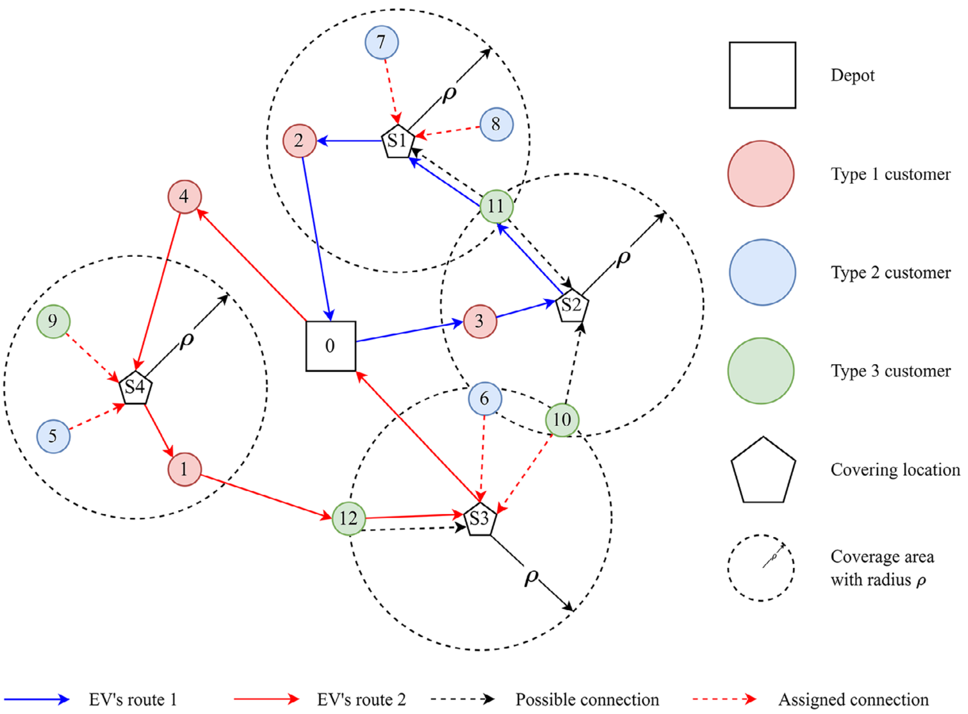

Fig. 1. An illustrative example of EVRPTW-PR-CL.

visited within a predefined time window $[e_i, l_i]$ with a service time s_i. It is noted that CLs typically serve multiple customers through centralized deliveries, allowing for more efficient service compared to direct customer visits. Accordingly, we follow Tilk et al. (2021) to assume that the service time at CLs is half that at customer nodes.

A fleet of m homogeneous EVs with load capacity of Q_V, battery capacity of B, charging rate of g, and consumption rate of r is available for each routing plan. Each used EV incurs a fixed charge of γ, and its route starts at the depot with a fully charged battery and ends at the depot.

Each customer $i \in N_C$ has a nonnegative demand d_i. Customer $i \in N_H \cup N_F$ can be served at their home location. For customer $i \in N_S \cup N_F$, their parcels will be collected at a compatible CL. Each customer i is compatible with a subset of replicated CLs, denoted by $S_i = \{p \in N_R | c_{ip} \leq \rho\}$, where ρ represents the maximum travel distance that customers accept to go to collect their parcels, denoting the coverage radius of each CL. Let $A_1 = \{(i, p) | i \in N_S \cup N_F, p \in N_R, \text{ and } c_{ip} \leq \rho\}$ denote all possible connections between customers $i \in N_S \cup N_F$ and replicated CLs $p \in N_L$. Each connection $(i, p) \in A_1$ corresponds with a connection cost δc_{ip} offered to customers, where δ denotes the compensation rate. We assume that CLs are to utilize existing facilities (e.g., convenience stores), thereby incurring no construction costs. While battery recharging costs are involved in travel costs, each CL used for serving customers incurs a fixed operational cost α to cover activities such as unloading, loading, and storage.

Figure 1 illustrates an example of EVRPTW-PR-CL, including a depot (marked by a square), four Type 1 customers (numbered 1–4 and marked by red circles 1–4), four Type 2 customers (numbered 5–8 and marked by blue circles), four Type 3 customers (numbered 9–12 and marked by green circles), and four CLs (mark by pentagons). Based on the coverage area of CLs (dash circles), possible connections between CLs and customers are represented by dashed lines. For example, CL S3 covers three customers: 6, 10, and 12. Note that a customer can be covered by several CLs (e.g., customer 10 can be assigned to S2 or S3). This example involves two used EVs (denoted by solid lines) and three CLs used to serve customers 5–10 (denoted by red dashed lines). All Type 1 customers are served by EVs, while all Type 2 customers are assigned to CLs. For Type 3 customers, customers 11 and 12 are served by EVs, but customers 9 and 10 are assigned to pick up their parcels at CLs. In addition, EV's route 1 can also visit CL S2 for recharging only.

For obtaining a EVRPTW-PR-CL solution, the model proposes decision variables—x_{ij}: 1 if arc $(i, j) \in A$ is traversed and 0 otherwise; o_{ij}: 1 if customer $i \in N_2$ collects parcels at CL $j \in N_L$ and 0 otherwise; z_i: 1 if CL $i \in N_L$ is used for serving customers and 0 otherwise; w_i: total parcels delivered to replicated CL $i \in N_R$; m_i: 1 if replicated CL $i \in N_R$ is used for serving customers and 0 otherwise; τ_i: service start time at node $i \in N$ for the EV arrival; u_i: current load upon the EV's arrival at node $i \in N$; y_i: battery state upon arriving at node $i \in N$; and Y_i: battery state upon departing from node $i \in N$:

$$\text{Minimize} \sum_{(i,j)\in A} c_{ij}x_{ij} + \delta \sum_{(i,j)\in A_1} c_{ij}o_{ij} + \gamma \sum_{(0,i)\in A} x_{0i} + \alpha \sum_{i\in N_L} z_i. \quad (1)$$

Objective function (1) minimizes four terms: total travel cost, connection costs, vehicle fixed cost, and operational costs of CLs:

$$\sum_{(i,j)\in A} x_{ij} + \sum_{(i,p)\in A_1} o_{ip} = 1 \qquad \forall\, i \in N_C \quad (2)$$

$$\sum_{(i,p)\in A_1} o_{ip} = 1 \qquad \forall\, i \in N_S \quad (3)$$

$$\sum_{(j,i)\in A} x_{ji} - \sum_{(i,j)\in A} x_{ij} = 0 \qquad \forall\, i \in N_1 \quad (4)$$

$$\sum_{(i,j)\in A} x_{ij} \le 1 \qquad \forall\, i \in N_R \quad (5)$$

$$\sum_{(0,i)\in A} x_{0i} \le m. \quad (6)$$

Constraints (2) and (3) enforce that each customer must be served by the associated delivery methods. Constraint (4) is the conservation flow constraint. Constraint (5) deals with arc connectivity related to CLs. Constraint (6) limits the number of used EVs:

$$\tau_i + t_{ij} + s_i m_i + g(Y_i - y_i) - l_0(1 - x_{ij}) \le \tau_j \qquad \forall\, i \in N_R, (i, j) \in A \quad (7)$$

$$\tau_i + t_{ij} + s_i - l_0(1 - x_{ij}) \le \tau_j \qquad \forall\, i \in N^+, (i, j) \in A \quad (8)$$

$$e_i \leq \tau_i \leq l_i \qquad \forall\, i \in N \tag{9}$$

$$u_j \leq u_i - d_i + Q_V(1 - x_{ij}) \qquad \forall\, i \in N_C, (i, j) \in A \tag{10}$$

$$u_j \leq u_i - w_i + Q_V(1 - x_{ij}) \qquad \forall\, i \in N_R, (i, j) \in A \tag{11}$$

$$u_i \leq Q_V \qquad \forall\, i \in N \tag{12}$$

$$y_j \leq y_i - rc_{ij}x_{ij} + B(1 - x_{ij}) \qquad \forall\, i \in N_C, (i, j) \in A \tag{13}$$

$$y_j \leq Y_i - rc_{ij}x_{ij} + B(1 - x_{ij}) \qquad \forall\, i \in N_3, (i, j) \in A \tag{14}$$

$$y_i \leq Y_i \leq B \qquad \forall\, i \in N_3. \tag{15}$$

Constraints (7) and (8) determine the service start time at each node. Constraint (9) enforces the visit time for each node to fall within its predefined time windows. Constraints (10) and (11) track the loading of an EV at each node. Constraint (12) ensures that the total load of an EV does not exceed its capacity. Constraints (13) and (14) track the remaining battery capacity when the EVs visit each node, respectively. Constraint (15) guarantees that the battery state does not exceed its capacity at CLs:

$$w_j = \sum_{(i,j)\in A_1} d_i o_{ij} \qquad \forall\, j \in N_R \tag{16}$$

$$\sum_{j\in\mathcal{K}_i} w_j \leq Q_L \qquad \forall\, i \in N_L \tag{17}$$

$$w_j \leq Q_L z_i \qquad \forall\, i \in N_L, j \in \mathcal{K}_i \tag{18}$$

$$Q_L \sum_{(i,j)\in A} x_{ij} \geq w_i \qquad \forall\, i \in N_R \tag{19}$$

$$Q_L m_i \geq w_i \qquad \forall\, i \in N_R \tag{20}$$

$$w_i \geq m_i \qquad \forall i \in N_R. \tag{21}$$

Constraint (16) calculates the total parcels required at each replicated CL and equals the total demand from customers picking up their parcels at that CL. Constraint (17) ensures that the total parcels delivered to each CL do not exceed the PL's capacity. Constraint (18) defines whether each CL is used for serving at least one customer or not. Constraints (19)–(21) ensure that a replicated CL must be visited by EVs if the customer's parcels are delivered to the replicated CL.

Algorithm 1. $VNS(\zeta^{shake}, \mathcal{S}, \zeta^{vnd}, \zeta^{ls}, \mathcal{N})$

1: $\sigma \leftarrow initializeSol()$; // see Section 4.3
2: $\sigma', \sigma^* \leftarrow \sigma, \sigma_f^* \leftarrow \emptyset$
3: **if** $isFeasible(\sigma)$ **then**
4: $\quad \sigma_f^* \leftarrow \sigma$
5: **end if**
6: $\iota \leftarrow 0, k \leftarrow 1, \eta^{shake} \leftarrow 1, \mathcal{R} \leftarrow \emptyset$
7: **while** $(\iota < \eta_{\max})$ **and** $\left(\iota - \iota_{imp} < \eta_{imp}\right)$ **do**
8: $\quad$ **if** $f_{gen}(\sigma) < f_{gen}(\sigma_f^*)$ **then**
9: $\quad\quad \sigma' \leftarrow \sigma$
10: $\quad$ **else**
11: $\quad\quad \sigma' \leftarrow \sigma_f^*$
12: $\quad$ **end if**
13: $\quad \sigma' \leftarrow shakeProcedure(\sigma', \mathcal{S}, \zeta^{shake}, \eta^{shake}, k)$; // see Section 4.4
14: $\quad \sigma' \leftarrow sequentialVND(\sigma', \mathcal{N}, \zeta^{vnd}, \zeta^{ls}, \iota)$; // see Section 4.5
15: $\quad \sigma' \leftarrow implementDP(\sigma')$; // see Section 4.6
16: $\quad$ **if** $f_{gen}(\sigma') < f_{gen}(\sigma)$ **then**
17: $\quad\quad \sigma \leftarrow \sigma'$
18: $\quad\quad k \leftarrow 1, \eta^{shake} \leftarrow 1, \iota_{imp} \leftarrow \iota$
19: $\quad$ **else**
20: $\quad\quad k \leftarrow k + 1, \eta^{shake} \leftarrow \max\left(\eta^{shake} + 1, \eta_{max}^{shake}\right)$
21: $\quad$ **end if**
22: $\quad$ **if** isFeasible(σ') **and** $f_{gen}(\sigma') < f_{gen}(\sigma_f^*)$ **then**
23: $\quad\quad \sigma_f^* \leftarrow \sigma'$
24: $\quad$ **end if**
25: $\quad updatePenalties()$; // see Section 4.2
26: $\quad \mathcal{R} \leftarrow \mathcal{R} \cup collectRoute(\sigma')$
27: $\quad$ **if modulo** $(\iota, \eta^{SPF}) = 0$ **then**
28: $\quad\quad \sigma' \leftarrow solveSPF(\mathcal{R})$ // see Section 4.7
29: $\quad$ **end if**
30: $\quad \iota \leftarrow \iota + 1$
31: **end while**
32: **return** σ_f^*

4. Solution method

A VNS algorithm was first introduced by Mladenović and Hansen (1997), and its framework is characterized by its simplicity, featuring only a few parameters. The concept of VNS has been recently applied to solve VRP variants such as EVRPTW (Çatay and Sadati, 2023), multidepot VRPTW (Bezerra et al., 2023), and covering delivery problem with EV and PLs (Vukićević et al., 2023). A comprehensive review of VNS applications up to 2023 is provided by Brimberg et al. (2023). In this research, we implement the VNS framework to solve the problem.

Algorithm 1 illustrates the procedure of VNS. Let σ denote the incumbent solution, σ' denote the temporary solution during the searching process, σ^* denote the best solution, and σ_f^* denote the best feasible solution. The algorithm starts with generating an initial solution by a construction algorithm (see Line 1). This solution is then assigned to be solutions σ' and σ^*, while it also becomes σ_f^* only if it is feasible (see Lines 2–4). Line 6 sets initial values of the current iteration ι, the neighborhood shaking operator k, the shaking level η^{shake}, and the column pool for SPF $\mathcal{R}$. A procedure is then iterated until implementing η_{max} iterations or η_{imp} iterations without improvement (see Line 7).

Regarding each iteration, the process begins by updating σ' (see Lines 8–11). Next, the shaking procedure is implemented to make perturbations to σ' based on predefined inputs (see Line 13). Line 14 executes the improvement procedure to enhance σ'. To further optimize σ', the dynamic programming (DP) procedure is employed to refine the feasible routes in σ' (see Line 15). If σ' is better than σ, then it replaces σ, and both η^{shake} and k are reset to 0, otherwise, η^{shake} and k are incremented by 1, up to a maximum value of η_{max}^{shake} (see Lines 16–20). The change of k represents the sequential neighborhood change step, while the change of η^{shake} follows Bezerra et al. (2023). Moreover, σ_f^* is updated whenever a better feasible solution is attained (see Lines 22 and 23). The penalty weights in the objective function are also updated (see Line 25). All nonoverlapping feasible routes within σ' are collected to store in the pool of columns $\mathcal{R}$ (see Line 26). Subsequently, the SPF is invoked in every η^{SP} iterations (see Lines 27 and 28).

The following sections detail the components of the proposed VNS approach. Section 4.1 describes the encoding of EVRPTW-PR-CL solution. Section 4.2 then shows how to evaluate the solution quality, which allows three types of violations (i.e., time windows, battery, and loading). Sections 4.3–4.5 discuss the main components in the proposed VNS. First, an initial solution is considered before implementing the VNS procedure (see Section 4.3). Second, the procedure is repeated in a sequence of three main steps: the shaking procedure (see Section 4.4), the improvement phase (see Section 4.5), and the neighborhood change step. Lastly, the integration of the classical VNS framework with DP (see Section 4.6) and a set partitioning formulation (SPF) (see Section 4.7) are presented.

4.1. Solution presentation

A EVRPTW-PR-CL solution is encoded by two 2D arrays described. Array 1 represents actual routes. Each row represents a route sequence that starts and ends at depot 0. Within each route, the EV serves Type 1 and 3 customers and visits some CLs to either drop off parcels and/or recharge batteries. We note that all Type 1 customers must be included in Array 1 to ensure the solution's feasibility. Array 2 represents customer assignments served at CLs. Its size is determined by the size of Array 1. Values at positions associated with customers from Array 1 are set to -1, indicating that no SP service during those visits. For positions associated with CLs from Array 1, values can represent either a Type 2 or Type 3 customer, or be set to -1 if the visit is *solely* for recharging batteries.

This representation ensures that all customers are served by their provided delivery options. Moreover, updating the connection costs, fixed costs for used EVs, and operational costs of CLs, as well as delivery options can be done in $\mathcal{O}(1)$ time complexity by adding/subtracting costs when inserting/removing a node. However, the solution may still have violations of time windows, battery consumption, and vehicle loading. To address them, penalties for these mentioned violations are incorporated into the objective function (see Section 4.2).

Array 1

EV route 1	0	3	S2	11	S1	S1	2	0	
EV route 2	0	4	S4	S4	1	12	S3	S3	0

Array 2

0	-1	-1	-1	8	7	-1	0	
0	-1	9	5	-1	-1	6	10	0

Fig. 2. Solution representation for the routing plan of Fig. 1.

The fixed operational cost of CLs is updated by tracking the total demand of customers assigned to each CL. The update follows two cases below.

- When customer i is assigned to CL j: If CL j is not used before assigning customer i (i.e., the current total demand at CL j is 0), then α is added to the objective function; otherwise, no additional cost is incurred. The total demand at CL j is updated by adding d_i.
- When customer i is removed from CL j: The total demand of CL j is first updated by subtracting d_i. After that, if the total demand becomes 0, indicating CL j is no longer in use, then α is subtracted from the objective function; otherwise, no further changes are made.

Figure 2 illustrates the solution representation for the example given in Fig. 1. In Array 1, two routes start and end at depot 0, visiting all Type 1 and some Type 3 customers (represented by white cells) as well as some CLs (represented by gray cells). Array 2 is then derived from Array 1, focusing only on gray cells. If a customer is assigned to a CL, then the corresponding value in Array 2 is set to that customer. For instance, customers 8 and 7 are assigned to CL S1 in route 1. For CLs visited solely for recharging, the corresponding value in Array is set to -1 (e.g., the position associated with CL S2 of route 1 in Array 1).

4.2. *Objective function with penalties*

Infeasible solutions in terms of load, battery, and time windows are considered to explore the search space of VNS. To ensure fairness with feasible solutions, penalty terms are added to the objective function (1), resulting in the generalized objective function $f_{gen}(\sigma)$ denoted as follows:

$$f_{gen}(\sigma) = \sum_{\kappa \in \sigma} f_{gen}(\kappa) = f(\sigma) + \sum_{\kappa \in \sigma} \lambda(\kappa). \tag{22}$$

Here, σ denotes a solution having n routes $\{\kappa_1, \ldots, \kappa_n\}$ with its objective function (1) denoted by $f(\sigma)$. Moreover, $\lambda(\kappa) = \rho_L \lambda_L(\kappa) + \rho_B \lambda_B(\kappa) + \rho_T \lambda_T(\kappa)$ represents the total penalty for route κ, comprising load penalty $\lambda_L(\kappa)$, time window penalty $\lambda_B(\kappa)$, and battery penalty $\lambda_T(\kappa)$, with each weighted by ρ_L, ρ_B, and ρ_T, respectively.

To enhance the exploration of infeasible solutions, dynamic updates to penalty weights are incorporated during the implementation of VNS. The weights are initialized as ρ_L^0, ρ_B^0, and ρ_T^0 and undergo adjustments within predefined ranges: $[\rho_L^{\min}, \rho_L^{\max}]$, $[\rho_B^{\min}, \rho_B^{\max}]$, and $[\rho_T^{\min}, \rho_T^{\max}]$.

Adopted from Goeke and Schneider (2015), the lower bounds for these weights are set to 0.5, while the upper bounds are determined by the initial objective values. These weights are then

Algorithm 2. A modified nearest neighborhood algorithm—*initializeSol*()

1: $\sigma \leftarrow \emptyset$
2: $\mathcal{U}_H \leftarrow N_H, \mathcal{U}_S \leftarrow N_S, \mathcal{U}_F \leftarrow N_F$
3: **while** $\mathcal{U}_H \neq \emptyset$ **do** {solve Type 1 customers}
4: $\quad$ $node, v, i \leftarrow greedySearch(\sigma)$
5: $\quad$ Insert *node* into Array 1 of σ at route v and position i
6: $\quad$ Insert -1 into Array 2 of σ at route v and position i
7: $\quad$ $\mathcal{U}_H \leftarrow \mathcal{U}_H \setminus \{node\}$
8: **end while**
9: **while** $\mathcal{U}_S \neq \emptyset$ **do** {solve Type 2 customers}
10: $\quad$ $node, v, i, locker \leftarrow greedySearch(\sigma)$
11: $\quad$ Insert *locker* into Array 1 of σ at route v and position i
12: $\quad$ Insert *node* into Array 2 of σ at route v and position i
13: $\quad$ $\mathcal{U}_S \leftarrow \mathcal{U}_S \setminus \{node\}$
14: **end while**
15: **while** $\mathcal{U}_F \neq \emptyset$ **do** {solve Type 3 customers}
16: $\quad$ $node, v, i, locker \leftarrow greedySearch(\sigma)$
17: $\quad$ **if** $locker = -1$ **then**
18: $\qquad$ Insert *node* into Array 1 of σ at route v and position i
19: $\qquad$ Insert -1 into Array 2 of σ at route v and position i
20: $\quad$ **else**
21: $\qquad$ Insert *locker* into Array 1 of σ at route v and position i
22: $\qquad$ Insert *node* into Array 2 of σ at route v and position i
23: $\quad$ **end if**
24: $\quad$ $\mathcal{U}_F \leftarrow \mathcal{U}_F \setminus \{node\}$
25: **end while**
26: $\sigma \leftarrow dynamicProgramming(\sigma)$ // see Section 4.6
27: **return**σ

updated in response to the feasibility of solutions. In case of consecutive violations (or satisfactions) of a constraint (either load, battery, or time windows) over $\eta^{penalty}$ iterations, the corresponding penalty weight is multiplied (or divided) by $\gamma^{penalty}$.

4.3. Construction of the initial solution

We propose a modified nearest neighborhood algorithm to generate an initial solution for VNS, including two stages. The first stage iteratively inserts customers into the solution σ with the lowest f_gen increment (Δf_{gen}). The second stage employs the DP procedure to potentially obtain optimal feasible routes (see Section 4.6). Algorithm 2 describes the procedure of *initializeSol*() in detail.

- Lines 1 and 2 initialize an empty solution σ and sets of unassigned customers $\mathcal{U}_H$, $\mathcal{U}_S$, and $\mathcal{U}_F$. These sets of unassigned customers are considered sequentially.
- First, Type 1 customers are iteratively inserted into σ (see Lines 3–7). At each iteration, the best insertion is defined by (*node*, v, i), where *node* denotes the selected customer, v denotes the route index, and i is the position index in route v (see Line 4). Next, *node* is inserted into σ at (v, i)

Algorithm 3. *shakeProcedure*($\sigma, \mathcal{S}, \zeta^{shake}, \eta^{shake}, \mathcal{S}_k$)

1: **for** $i = 1; i \leq \eta^{shake}; i++$ **do**
2:     **if** $\zeta^{shake} =$ "*sequential*" **then**
3:         $\mathcal{S}_k$ is predefined during the procedure of VNS
4:         $\sigma \leftarrow \mathcal{S}_k(\sigma)$
5:     **else if** $\zeta^{shake} =$ "*random*" **then**
6:         $\mathcal{S}_k \leftarrow$ select random from $\mathcal{S}$
7:         $\sigma \leftarrow \mathcal{S}_k(\sigma)$
8:     **else if** $\zeta^{shake} =$ "*adaptive*" **then**
9:         $\mathcal{S}_k \leftarrow$ *rouletteWheel*($\mathcal{S}$)
10:         $\sigma \leftarrow \mathcal{S}_k(\sigma)$
11:     **end if**
12: **end for**
13: **return** σ

in Array 1 (see Line 5), while -1 is set at the associated position in Array 2 (see Line 6). This procedure terminates when all Type 1 customers are served.

- Second, Type 2 customers must be assigned to one of their possible CLs (see Lines 9–13). At each iteration, the best insertion is defined as (*node*, *v*, *i*, *locker*), where *locker* is the selected CL to serve the customer *node* (see Line 10). Next, *locker* is inserted into σ at (v, i) in Array 1 (see Line 11) while *node* is assigned to the associated position in Array 2 (see Line 12). This procedure is repeated until all Type 2 customers are assigned.
- Third, Types 3 customers can be served by either EVs or CLs (see Lines 15–24). Similarly, the best insertion is generalized as (*node*, *v*, *i*, *locker*). If the best insertion resembles the case used for Type 1 customer insertion, then the value of *locker* is set to -1 (see Lines 17–19); otherwise, the best insertion is similar to the case of Type 2 customer insertion (see Lines 20–22).
- *greedySearch*(σ) is implemented to find the best insertions. Particularly, each unassigned customer is considered and tested for all possible insertions in the incomplete solution σ. Creating a new route is also considered as a possible insertion if the number of used EVs is less than the available number of EVs. The best insertion is based on the lowest cost increment $\Delta f_{gen} = f_{gen}^{after} - f_{gen}^{before}$.
- Finally, the DP procedure is implemented (see Line 26).

4.4. Shaking procedure

We denote *shakeProcedure*($\sigma, \mathcal{S}, \zeta^{shake}, \eta^{shake}, \mathcal{S}_k$) as the shaking procedure, aiming to perturb the solution σ using a set of shaking operators $\mathcal{S} = \{\mathcal{S}_1, \ldots, \mathcal{S}_8\}$ to escape from local optima traps. Algorithm 3 describes this procedure with two main ideas to enhance VNS: the level of shake and the selection of shaking operators. First, η^{shake}, defined during the VNS procedure, represents the level of shake, which is the number of times that shaking operators are applied to σ at each iteration (Bezerra et al., 2023) (see Line 1). Next, ζ^{shake} denotes the strategy for selecting the shaking operators, including three strategies defined as follows.

- With *sequential* selection (see Lines 2–4): $\mathcal{S}_k$ is predefined at each iteration in VNS (Algorithm 1), which is implemented to shake σ η^{shake} times.
- With *random* selection (see Lines 5–7): a shaking operator $\mathcal{S}_k$ is randomly chosen and applied to shake σ. This procedure is repeated η^{shake} times.
- With *adaptive* selection (see Lines 8–11): a shaking operator $\mathcal{S}_k$ is selected based on the probability associated with its performance in VNS, denoted by $rouletteWheel()$. The solution σ is then shaken by the selected shaking operator η^{shake} times.

Regarding the set of shaking operators, these operators are categorized into two types: node-based operators (see Section 4.4.1) and option-based operators (see Section 4.4.2).

4.4.1. Node-based shaking operators

We employ seven node-based shaking operators, denoted by $\mathcal{S}_N = \{\mathcal{S}_1, \ldots, \mathcal{S}_7\}$. These operators are applied to Array 1 of σ, while the associated cells in Array 2 of σ remain unchanged. Creating a new route is also a possible insertion considered in all operators if the number of used EVs ($usedEV$) is less than the number of available EVs (m). Specifically, these operators are presented as follows.

- *Random_remove_destroy* move (denoted by $\mathcal{S}_1(\sigma)$) (Ropke and Pisinger, 2006): this move randomly removes $\eta = rand(1, \eta^{rem})$ nodes in Array 1 and then reinserts them into σ. We define $\eta^{rem} = 0.1|N_C|$.
- *Eliminate_shortest_routes* move (denoted by $\mathcal{S}_2(\sigma)$) (Yu et al., 2023a): this move eliminates the route with the fewest nodes and then reinserts the removed nodes (i.e., customers and CLs serving customers) into σ. If $usedEV = 1$, then $\mathcal{S}_3$ is implemented instead.
- *Split_longest_routes* move (denoted by $\mathcal{S}_3(\sigma)$) (Yu et al., 2023a): this move separates the route with the most nodes into two new routes. The splitting point is randomly selected. If $usedEV = m$, then $\mathcal{S}_4$ is employed instead.
- *2-Opt** move (denoted by $\mathcal{S}_4(\sigma)$): this move selects two random routes, and each route removes an edge. These edges are then reconnected to create two new routes. If $usedEV = 1$, then $\mathcal{S}_3$ is implemented instead.
- *Eliminate_worst_routes* move and *Split_worst_route* move (denoted by $\mathcal{S}_6(\sigma)$ and $\mathcal{S}_7(\sigma)$) (Yu et al., 2023a): these moves are similar to $\mathcal{S}_2$ and $\mathcal{S}_3$, respectively, where the selected route is the one with the highest total penalty costs.

4.4.2. Option-based shaking operators

With the option-based idea, we employ *Change_delivery_option* move (denoted by $\mathcal{S}_8(\sigma)$). To modify σ, $\mathcal{S}_8(\sigma)$ randomly selects a *node* in Array 1, which must be either a Type 3 customer or a CL serving customers. Next, its delivery option is either retained or changed with equal probability. If the decision is to change the delivery option, then the process runs as follows.

- If *node* is the Type 3 customer, then its delivery option is changed to SP service. Particularly, *node* is removed and replaced by a CL, which is randomly selected from the set of its compatible CLs in Array 1. Next, *node* is inserted into the associated position in Array 2.

Algorithm 4. *sequentialVND*$(\sigma, \mathcal{N}, \zeta^{vnd}, \zeta^{ls}, \iota)$

if $\zeta^{vnd} =$ "*random*" **then**
  $\mathcal{N}^{seq} \leftarrow$ Randomly shuffle the order of set $\mathcal{N}$
else if $\zeta^{vnd} =$ "*adaptive*" **then**
  $\mathcal{N}^{seq} \leftarrow$ Update the order of $\mathcal{N}$ by their performance
end if
repeat
  if $\zeta^{ls} =$ "*first*" **then**
    $\sigma' \leftarrow$ the first better neighbor of $\mathcal{N}_k(\sigma)$
  else if $\zeta^{ls} =$ "*best*" **then**
    $\sigma' \leftarrow$ the best neighbor of $\mathcal{N}_k(\sigma)$
  else if $\zeta^{ls} =$ "*hybrid*" **then**
    if modulo $(\iota, \eta^{hybrid}) = 0$ **then**
      $\sigma' \leftarrow$ the best neighbor of $\mathcal{N}_k(\sigma)$
    else
      $\sigma' \leftarrow$ the first better neighbor of $\mathcal{N}_k(\sigma)$
    end if
  end if
  if $f_{gen}(\sigma') < f_{gen}(\sigma)$ **then**
    $k \leftarrow 1$
  else
    $k \leftarrow k + 1$
  end if
until $k = |\mathcal{N}|$
return σ

- If *node* is the CL associated with a customer *cus* in Array 2, then there are two cases. First, if *cus* is the Type 2 customer, then *node* is replaced by another compatible CL of *cus*. Second, if *cus* is the Type 3 customer, then the solution is changed by either replacing *node* with another compatible CL of *cus* or removing *node* from Array 1. In the latter case, *node* is replaced by *cus* in Array 1, and the associated position in Array 2 becomes −1.

4.5. Improvement procedure

A sequential variable neighborhood descent (VND) is employed in the improvement phase, denoted by *sequentialVND*$(\sigma, \mathcal{N}, \zeta^{vnd}, \zeta^{ls}, \iota)$. Here, σ represents the considered solution, $\mathcal{N}$ is a set of neighborhood operators, ζ^{vnd} determines the strategy for ordering operators, ζ^{ls} specifies the improvement strategy, and ι is the iteration index. In this study the set of neighborhood operators includes $\mathcal{N} = \{\mathcal{N}_1, \ldots, \mathcal{N}_{11}\}$. These operators are categorized into three types: node-based operators $\mathcal{N}_N = \{\mathcal{N}_1, \ldots, \mathcal{N}_8\}$ (see Section 4.5.1), charging-based operators $\mathcal{N}_C = \{\mathcal{N}_9, \mathcal{N}_{10}\}$ (see Section 4.5.2), and an option-based operator $\mathcal{N}_{11}$ (see Section 4.5.3).

Algorithm 4 details the procedure of *sequentialVND*$(\sigma, \mathcal{N}, \zeta^{vnd}, \zeta^{ls}, \iota)$. Initially, $\mathcal{N}^{seq}$ is defined as a reordered sequence of $\mathcal{N}$ based on the ζ^{vnd} strategy. Three ordering strategies are proposed. For *fixed* strategy, the sequential order is set as $\mathcal{N}^{seq} = \{\mathcal{N}_8, \mathcal{N}_7, \mathcal{N}_5, \mathcal{N}_6, \mathcal{N}_9, \mathcal{N}_4, \mathcal{N}_3, \mathcal{N}_1, \mathcal{N}_2, \mathcal{N}_{10}, \mathcal{N}_{11}\}$.

For *random* strategy, the sequence is shuffled (see Lines 1 and 2). For *adaptive* strategy, the order is sorted based on their performance scores (see Lines 3 and 4). Following Bezerra et al. (2023), improving σ with neighborhood move $\mathcal{N}_k$ increases the score of $\mathcal{N}_k$ by 5. However, if a new σ_f^* is obtained, then the score of $\mathcal{N}_k$ is increased by 15.

Each neighborhood move in $\mathcal{N}^{seq}$ is sequentially explored, searching through all the possible solutions associated with it. Afterward, the improved solution is obtained by using one of three improvement strategies: (a) for the *first* improvement strategy, the process terminates upon finding a better neighbor solution (see Lines 7 and 8); (b) for the *best* improvement strategy, all neighbor solutions are evaluated, and the best one is selected (see Lines 9 and 10); and (c) for the *hybrid* strategy, the first strategy is mainly used, while the best strategy is applied after every $\eta^{hybrid} = 10$ iterations (see Lines 11–15).

If a better solution than σ is found, then we return to the first neighborhood move (see Lines 18 and 19); otherwise, the next neighborhood move is used (see Lines 20 and 21). The sequence ends when all neighborhood moves are implemented without any improvement (see Line 24).

4.5.1. Node-based operators

This section describes eight node-based neighborhood moves widely implemented in research (Subramanian et al., 2010). All node-based operators change only the sequence of routes (Array 1), while the assignments (Array 2) are unchanged. There are four intraroute moves (considering one route) and four interroute moves (considering two different routes) in a neighborhood, which are described as follows.

- $\mathcal{N}_1(\sigma)$: *Swap(1,1)* generates a neighborhood solution by swapping two customers.
- $\mathcal{N}_2(\sigma)$: *Shift(1,0)* removes a node and then reinserts it into another position.
- $\mathcal{N}_3(\sigma)$: *Shift(2,0)* removes two consecutive nodes and then reinserts them into another position.
- $\mathcal{N}_4(\sigma)$: *2-Opt* selects two points in a route, reversing the segment between these points.
- $\mathcal{N}_5(\sigma)$: *Swap(1,1)-Interroute* is similar to $\mathcal{N}_1(\sigma)$, but it swaps two nodes in different routes.
- $\mathcal{N}_6(\sigma)$ and $\mathcal{N}_7(\sigma)$: *Shift(1,0)-Interroute* and *Shift(2,0)-Interroute* are similar to $\mathcal{N}_2(\sigma)$ and $\mathcal{N}_3(\sigma)$, respectively. However, the new insertion position is a different route.
- $\mathcal{N}_8(\sigma)$: 2-Opt* removes two edges (i_1, j_1) and (i_2, j_2) in two different routes. Later, i_1 (resp., i_2) is connected to j_2 (resp., j_1) to generate two new routes.

4.5.2. Charging-based operator

Two charging-based operators are adopted: *add_CL* (denoted by $\mathcal{N}_9(\sigma)$) and *remove_CL* (denoted by $\mathcal{N}_{10}(\sigma)$). The former operator attempts to reduce the battery penalty by inserting CLs for recharging EV batteries, while the latter one is implemented to remove unnecessary CLs (Cortés-Murcia et al., 2019).

4.5.3. Option-based operator

To address the delivery options, we introduce a new neighborhood operator, named *Change_option* (denoted by $\mathcal{N}_{11}(\sigma)$). This move considers changing the delivery option of a customer.

4.6. Dynamic programming

A DP procedure, denoted by *implementDP*(σ), is implemented to optimally insert CLs for battery recharge, preventing battery depletion. The detailed procedure of DP appears in Schiffer and Walther (2018). Due to the significant computational time required to perform DP, we create a hashmap containing pairs of routes before and after the implementation of DP. Before implementing DP, we therefore need to check whether the currently evaluated route has been processed by DP previously. If such a route is found in the hashmap, then the paired route consisting of optimal CLs, placement is directly adopted; otherwise, DP is performed on the route, and a new pair associated with the route is added to the hashmap.

4.7. Set partitioning problem

The proposed VNS also utilizes a set partitioning formulation (SPF) to further improve the final solution. SPF has been successfully implemented in EVRP variants (Cortés-Murcia et al., 2019). Let $\mathcal{R}$ denote a set of feasible nonduplicated routes, which is collected during the implementation of VNS. For each route $k \in |\mathcal{R}|$, the route's cost, denoted by c_k^{col}, includes travel cost, the fixed cost for used EVs, and the connection cost. Moreover, o_{ik} indicates whether the route k visits customer $i \in N_C$, and p_{ik} indicates whether the CL $i \in N_L$ is used for serving customers at route k.

The implementation of SPF is denoted by *solveSPF*($\mathcal{R}$), where $\mathcal{R}$ denotes the input obtained from all feasible routes when seeking neighborhood solutions. The result is a feasible solution σ, determined by the following decision variables. Let θ_k denote a binary variable to indicate whether route $k \in |\mathcal{R}|$ is selected as part of the final solution. Moreover, variable z_i is reused for determining whether the CL $i \in N_L$ is used for serving customers or not. The proposed SPF (denoted by F_{SPF}) is formulated as follows:

$$(F_{SPF}) \quad \text{Minimize} \sum_{k \in \mathcal{R}} c_k^{col} \theta_k + \alpha \sum_{i \in N_L} z_i \tag{23}$$

$$\sum_{k \in |\mathcal{R}|} o_{ik} \theta_k = 1 \qquad \forall\, i \in N_H \cup N_F \tag{24}$$

$$\sum_{k \in |\mathcal{R}|} o_{ik} \theta_k \geq 1 \qquad \forall\, i \in N_S \tag{25}$$

$$\sum_{k \in |\mathcal{R}|} p_{ik} \theta_k \leq |\mathcal{R}| z_i \qquad \forall\, i \in N_L \tag{26}$$

$$\sum_{k \in |\mathcal{R}|} \theta_k \leq m \tag{27}$$

$$\theta_k, z_i \in \{0, 1\} \qquad \forall\, k \in |\mathcal{R}|, i \in N_L. \tag{28}$$

Objective function (23) is to minimize two terms. The first term represents the travel cost, the EV fixed cost, and the connection costs. The second term is the compensation costs. Constraint

(24) guarantees that Type 1 and 3 customers are served exactly one time. However, we relax the SPF by allowing each Type 2 customer to be served more than one time, represented by Constraint (25). Note that after a solution is obtained, unnecessary assignments of Type 2 customers need to be removed. Constraint (26) records which CLs are used for serving customers. Constraint (27) limits the available number of EVs. Constraint (28) defines the ranges of decision variables. Due to delivery options ensured in the collected feasible routes, we do not need to consider them in the SPF.

5. Computational results

This section conducts numerical experiments to extensively highlight our contributions. Section 5.1 first explains how we generate EVRPTW-PR-CL instances. Section 5.2 presents the tuning of parameters used in VNS to improve the solution quality. Section 5.3 presents the performance of the proposed VNS against state-of-the-art algorithms for solving EVRPTW-PR. Section 5.4 then provides the results of EVRPTW-PR-CL instances obtained by implementing both the proposed mathematical model and VNS. While the mathematical model in Section 3 is solved using the GUROBI solver, VNS is run in C++ Microsoft Visual Studio 2022. All experiments are executed on a computer with an Intel® Core™ i7-9700 CPU @ 3.0 GHz.

5.1. Description of the test instances

We generate EVRPTW-PR-CL benchmark instances based on existing EVRPTW instances proposed by Schneider et al. (2014). These include 36 small-sized instances (i.e., 12 instances with 5, 10, and 15 customers) with 3–8 CSs and 56 large-sized instances with 100 customers with 21 CSs. To address CLs and delivery options in EVRPTW-PR-CL instances, adoptions and modifications are highlighted as follows.

- The properties of EVs are adopted by Schneider et al. (2014).
- Customers are randomly categorized into one of three customer types (i.e., Types 1–3). It is assumed that the distribution of customers among these types is equal, with an equal number assigned to each type.
- The locations of CSs in EVRPTW instances are utilized as the locations of CLs in EVRPTW-PR-CL instances.
- Time windows for customers served at their private locations (AHD method) are taken by the customer time windows in EVRPTW instances, while the time windows for CLs are set as $[0, b_0]$ (i.e., the entire time horizon).
- Based on Vukićević et al. (2023), the covering radius ρ is defined by

$$\rho = d_{\min} + \chi \frac{d_{\max} - d_{\min}}{4}.$$

Herein, d_{max} and d_{min} represent the maximum and minimum travel distances between customers and CLs. χ denotes a covering level, initially set to 2. If a customer cannot be compatible with

Table 1
Summary of parameter descriptions and their values

Parameter	Description	Value
η_{max}	Maximum iterations implemented in VNS	1000[a]
η_{max}^{noimp}	Maximum consecutive iterations without improvement	100[a]
η^{SPF}	Number of iterations for implementing *solveSPF*()	50
η_{max}^{shake}	Maximum shaking level	5[b]
$\eta^{penalty}$	Number of consecutive iterations to obtain feasible (infeasible) solutions for implementing *updatePenalties*()	10[b]
$\gamma^{penalty}$	Reaction factor *updatePenalties*()	1.05[b]
$\rho_{tw}^{min}, \rho_{tw}^{max}, \rho_{tw}^{0}$	Time window penalty's weights	See Section 4.2
$\rho_{ba}^{min}, \rho_{ba}^{max}, \rho_{ba}^{0}$	Battery penalty's weights	See Section 4.2
$\rho_{lo}^{min}, \rho_{lo}^{max}, \rho_{lo}^{0}$	Load penalty's weights	See Section 4.2
ζ^{shake}	Shaking strategies	*sequential*[b]
ζ^{vnd}	Sequential order strategies	*random*[b]
ζ^{ls}	Improvement strategies	*first*[b]

[a] η^{max} and η_{max}^{noimp} are set to 1000 and 100, respectively, to ensure reasonable computation time.
[b] The final tuned values.

any CL within the covering radius, then the nearest CL to the customer is selected to cover them. The impact of the covering radius is further analyzed by varying the covering level within a range $\chi = \{1, 2, 3, 4\}$ (see Section 6.2).

- Regarding the connection cost, we vary the compensation factor within a range $\delta = \{0, 0.5, 1.5, 2, 5, 10, 15\}$. This parameter will be analyzed in Section 6.3.
- The fixed costs of used EVs and CLs are set to 100 and 50, respectively. We assume that the CL's capacity equals the EV's capacity.
- Based on EVRPTW instances, customers served by the AHD method have a constant service time. For CLs, if an EV drops off parcels at CLs, then a CL's service time is half of the customer's service time. However, if an EV visits the CLs only to charge its battery, no service time is included.

5.2. Parameter setting for VNS

To select strategies and tune parameters, we adopt the one-factor at time (OFAT) approach, executing VNS 10 times for each large-sized instance of EVRPTW-PR. The OFAT method involves iteratively testing individual candidate values for a parameter while keeping other parameters at one value. This problem first focuses on tuning the strategies employed in the shaking and improvement procedures (i.e., ζ^{shake}, ζ^{vnd}, and ζ^{ls}). Following this, three parameters (i.e., η_{max}^{shake}, $\eta^{penalty}$, and $\gamma^{penalty}$) are tuned, while the remaining parameters are adopted from the literature. To ensure a fair comparison with existing algorithms, we impose limits on the number of iterations to maintain reasonable computation times (i.e., $\eta^{max} = 1000$ and $\eta_{max}^{noimp} = 100$). All strategies and parameters are listed in Table 1.

Table A1 provides a summary of the tuning results for the strategies used in components. Strategies ζ^{shake}, ζ^{vnd}, and ζ^{ls} are sequentially tuned, with each component having three possible strategies. Initial strategies are set as Combination 1 in the table (i.e., $\zeta^{shake} =$ "*sequential*", $\zeta^{vnd} =$ "*random*″, and $\zeta^{ls} =$ "$first$"). For each tuned component, the best combination is selected based on the smallest average objective value ($\bar{f}$). Finally, VNS is implemented with the settings of Combination 7 in the table for solving subsequent experiments.

Table A2 presents the results of tuning parameters, including their initial values and final values. Each parameter has three candidate values for tuning, and the tuning order aligns with the reporting sequence in the table. The selection of the best value for each parameter is also determined based on the best objective value ($\bar{f}$).

5.3. Computational results on EVRPTW-PR instances

The proposed VNS is first implemented to the existing benchmark EVRPTW-PR instances, which involves executing the approach 10 times for each instance. While VNS obtains optimal solutions (proven by Keskin and Çatay (2016)) for all small-sized instances, the results of VNS on solving large-sized instances are compared against those obtained from the state-of-the-art algorithms proposed by Keskin and Çatay (2016) (KÇ), Schiffer and Walther (2018) (SW), Hiermann et al. (2019) (HHPV), Cortés-Murcia et al. (2019) (CPA), Yu et al. (2023b) (YAC), Erdelić and Carić (2022) (EC), and Çatay and Sadati (2023) (ÇS).

Table A3 presents the detailed results for solving the large-sized instances. Each algorithm's best solution from 10 runs is represented as f_b for each instance. The results of VNS also involve the average solution (f_a) and the average computational time ($\bar{t}$) in minutes. The gap between the average and the best solutions ($\Delta f_{(a-b)}[\%]$) shows the VNS stability (i.e., $\Delta f_{(a-b)} = 0.62\%$). We then compare the results of VNS with those of the remaining algorithms by determining the gaps among them ($\Delta f[\%] = \frac{f_b^{VNS} - f_b^{alg}}{f_b^{VNS}}$).

Table 2 summarizes that VNS outperforms the state-of-the-art algorithms, achieving an average improvement in solutions ranging from 0.03% to 1.55%. Notably, it provides 33 BKSs, with 27 out of them being new BKSs. Specifically, VNS provides better results than other algorithms for all subsets (i.e., C-1, C-2, R-1, R-2, RC-1, and RC-2). However, for subsets R-2 and RC-1, our results are worse than those of KÇ and SW on average, due to their utilization of more EVs versus our approach.

To sum up, VNS shows excellent performance in solving EVRPTW-PR instances within a reasonable computation time; that is, 2.55 minutes compared to 2.56 up to 16.77 minutes of state-of-the-art algorithms. These results imply a high possibility for addressing EVRPTW-PR-CL instances in Section 5.4.

5.4. Computational results on EVRPTW-PR-CL instances

This section presents the performance of VNS against GUROBI using EVRPTW-PR-CL instances under specific settings: $\delta = 1$, $\gamma = 100$, $\alpha = 50$, and $\chi = 2$. A time limit of two hours is imposed on

Table 2
Comparison of the results of the large-sized EVRPTW-PR among algorithms

Subset	Average of the best solution of 10 runs (f_b)								Average gap ($\Delta\bar{f}$[%])						
	KÇ	SW	HHPV	CPA	YAC	EC	ÇS	VNS	KÇ	SW	HHPV	CPA	YAC	EC	ÇS
C-1	1004.04	996.76	1003.34	994.57	1005.62	1018.52	1004.13	994.52	−0.94	−0.23	−0.92	−0.01	−1.17	−2.37	−0.94
C-2	629.95	629.82	640.99	629.82	629.82	629.82	629.95	629.82	−0.02	0.00	−1.77	0.00	0.00	0.00	−0.02
R-1	1221.70	1202.15	1239.72	1200.95	1222.96	1206.17	1209.25	1200.67	−1.77	−0.17	−3.29	−0.03	−1.87	−0.43	−0.78
R-2	908.89[a]	910.04[a]	914.76	912.16	918.90	914.50	914.00	912.23	0.38	0.17	−0.38	−0.09	−0.78	−0.35	−0.29
RC-1	1394.11	1359.97[a]	1389.71	1366.07	1404.22	1376.68	1375.97	1367.10	−1.91	0.49	−1.56	0.05	−2.67	−0.70	−0.62
RC-2	1140.07	1134.66	1142.61	1133.29	1142.23	1141.25	1138.19	1130.52	−0.87	−0.39	−1.00	−0.31	−1.07	−1.01	−0.70
Average	1049.79	1038.90	1055.19	1039.48	1053.96	1047.82	1045.25	1039.14	−0.86	−0.03	−1.55	−0.06	−1.27	−0.78	−0.57

[a]Some solutions use more EVs than our solutions.

Table 3
Summary of the comparison of the results obtained from three versions of VNS

Subset	VNS-noOption (1)			VNS-noSPF (2)			VNS (3)			GAP (Δf[%])			
	f_b	f_a	$\bar{t}$ (s)	f_b	f_a	$\bar{t}$ (s)	f_b	f_a	$\bar{t}$ (s)	Δf_b (1)	Δf_b (2)	Δf_a (1)	Δf_a (2)
5C	386.33	386.33	0.05	365.19	365.19	0.02	365.19	365.19	0.07	−5.79	0.00	−5.79	0.00
10C	591.64	591.64	0.11	523.38	523.60	0.09	523.27	524.91	0.15	−13.07	−0.02	−12.71	0.25
15C	701.06	701.47	0.30	647.34	649.16	0.25	647.01	648.86	0.41	−8.35	−0.05	−8.11	−0.05
C-1	3218.31	3226.51	17.67	2607.68	2650.80	68.24	2598.32	2635.62	225.42	−23.86	−0.36	−22.42	−0.58
C-2	1916.64	1928.83	33.89	1635.39	1666.20	118.28	1631.20	1660.46	121.41	−17.50	−0.26	−16.16	−0.35
R-1	2746.32	2763.41	15.99	2359.19	2412.91	65.21	2345.70	2397.37	76.47	−17.08	−0.58	−15.27	−0.65
R-2	1903.74	1922.80	40.29	1515.93	1555.90	236.31	1515.93	1555.37	235.51	−25.58	0.00	−23.62	−0.03
RC-1	3238.08	3286.14	15.13	2796.07	2847.25	61.39	2775.80	2821.38	71.98	−16.65	−0.73	−16.47	−0.92
RC-2	2417.96	2435.37	36.89	1852.20	1903.74	163.53	1851.96	1900.26	161.44	−30.56	−0.01	−28.16	−0.18

the GUROBI solver. Furthermore, the performances of the proposed option-based operators (i.e., $\mathcal{N}_{11}$ and $\mathcal{S}_8$) and the SPF in VNS are evaluated by considering two alternative variants of VNS: (1) the first variant "VNS-noOption" represents VNS without option-based operators, where $\mathcal{S}_8$ and $\mathcal{N}_{11}$ are excluded from VNS; and (2) the second variant "VNS-noSPF" is VNS without the implementation of SPF (i.e., Lines 26–28 in Algorithm 1 are omitted). In summary, each instance is solved using GUROBI, VNS, and two alternative variants of VNS.

Tables A4 and A5 present detailed results for small-sized instances (i.e., 5, 10, and 15 customers) and large-sized instances (i.e., 100 customers), respectively. The information includes the best solutions (f_b), average solutions (f_a), results obtained by GUROBI (f), gaps between the best and average solutions ($\Delta f_{(a-b)}$), and average computational time ($\bar{t}$) in seconds. Results from VNS are then compared against other approaches by the gaps between their best solutions (Δf[%]). For the small-sized instances, GUROBI yields feasible solutions for all instances with 21 optimal solutions, while VNS achieves the same or superior solutions with an improvement of 0.16% and significantly reduced computation time (i.e., 0.21 seconds on average). For the large-sized instances, VNS not only outperforms GUROBI by 5.36% in terms of solution quality, but also obtains this much faster (i.e., 149.57 seconds on average).

Table 3 provides a summary comparison of results obtained from three variants of VNS: VNS-noOption (1), VNS-noSPF (2), and VNS (3). This shows the impacts of option-based operators and the SPF in the proposed VNS, highlighted as follows.

- The option-based operators significantly improve in the best and average solutions across all subsets, ranging from 5.79% to 30.56% and from 5.79% to 28.16%, respectively. However, it is worth noting that this improvement comes at a reasonable cost in terms of computational time.
- The implementation of SPF slightly improves the solution quality for large-sized subsets (i.e., from 0.01% to 0.73%), albeit with a slight increase in computation time.

In summary, the proposed algorithm effectively addresses EVRPTW-PR-CL by integrating optional-based options and applying the SPF.

Table 4
Comparison of the cost components among three scenarios of EVRPTW-PR-CL instances

Instance	Scenario 1					Scenario 2					Scenario 3				
	f	$f^{(1)}$	$f^{(2)}$	$f^{(3)}$	$f^{(4)}$	f	$f^{(1)}$	$f^{(2)}$	$f^{(3)}$	$f^{(4)}$	f	$f^{(1)}$	$f^{(2)}$	$f^{(3)}$	$f^{(4)}$
C103C15	648.46	348.46	0	0	300	687.16	267.93	69.23	150	200	577.13	228.14	98.99	50	200
C106C15	575.13	275.13	0	0	300	651.61	228.44	123.17	100	200	473.31	149.10	174.21	50	100
C202C15	583.62	383.62	0	0	200	644.32	207.46	186.86	150	100	546.74	235.44	111.30	100	100
C208C15	500.55	300.55	0	0	200	663.70	274.38	189.32	100	100	458.72	274.18	34.54	50	100
R102C15	912.78	412.78	0	0	500	680.87	225.04	105.83	150	200	561.00	160.17	200.83	100	100
R105C15	736.15	336.15	0	0	400	654.55	183.93	120.62	150	200	574.23	218.62	105.62	50	200
R202C15	558	358	0	0	200	547.53	200.95	146.58	100	100	468.87	178.44	140.43	50	100
R209C15	413.24	313.24	0	0	100	578.50	244.72	83.78	150	100	413.24	313.24	0.00	0	100
RC103C15	797.67	397.67	0	0	400	778.32	259.51	168.81	150	200	709.21	244.64	114.57	150	200
RC108C15	670.25	370.25	0	0	300	730.41	252.75	127.66	150	200	595.57	291.71	53.87	50	200
RC202C15	594.39	394.39	0	0	200	595.48	312.71	82.77	100	100	552.18	362.46	39.72	50	100
RC204C15	503.38	403.38	0	0	100	555.62	208.23	147.38	100	100	438.01	262.32	25.69	50	100
Average	624.47	357.80	0.00	0.00	266.67	647.34	238.84	129.33	129.17	150	530.68	243.20	91.65	62.50	133.33

6. Managerial insights

Additional experiments are conducted to study the effects of employing CLs in the proposed problem. First, the trade-off between cost components in EVRPTW-PR-CL is presented in Section 6.1. Second, Section 6.2 discusses the sensitivity of the compensation rate, while the impact of the covering level is presented in Section 6.3. We utilize the EVRPTW-PR-CL instances with 15 customers and use their best routing plans obtained by executing 10 VNS runs.

6.1. Trade-off analysis

To analyze the trade-off of cost components (i.e., travel cost, connection cost, the used CLs' cost, and the used EVs' cost), three different scenarios are considered with varying proportions of customer types as follows.

- Scenario 1: All customers are Type 1, which implies that CLs are disregarded (EVRPTW-PR).
- Scenario 2: Customer types are equally distributed; that is, EVRPTW-PR-CL (see Section 5.1).
- Scenario 3: All customers are Type 3 customers to access how customer types influence routing plans.

Table 4 reports the detailed results with cost components. Column f represents the overall cost, column $f^{(1)}$ denotes the travel cost, column $f^{(2)}$ denotes the connection costs, column $f^{(3)}$ represents the fixed cost of used CLs, and column $f^{(4)}$ represents the fixed cost of used EVs. Observations are drawn as follows.

- A trade-off between cost components is observed between Scenarios 1 and 2. While Scenario 1 offers a better overall cost than Scenario 2 (due to connection cost ($f^{(2)}$) and the fixed cost of used

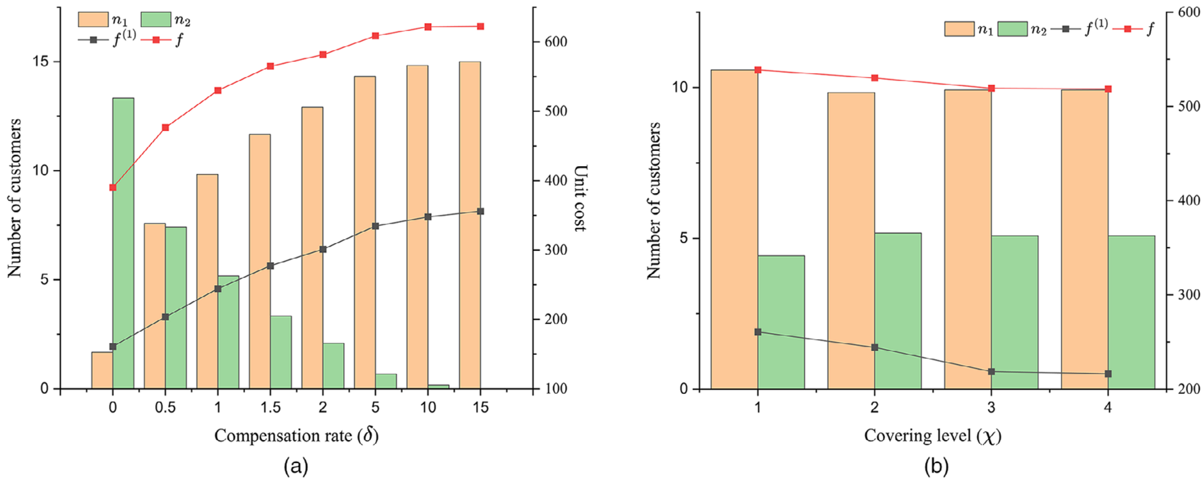

Fig. 3. Effects of (a) compensation rate and (b) covering level in EVRPTW-PR-CL.

CLs ($f^{(3)}$)), it incurs higher travel costs ($f^{(1)}$) and a greater number of used EVs ($f^{(4)}$), resulting in negative environmental impacts.

- The more customers are willing to be served by either Option 1 or Option 2, the better are the routing plans in terms of the overall cost that are provided. Indeed, Scenario 3 provides better routing plans than Scenario 2.
- The use of CLs for serving customers can significantly reduce the overall cost, especially travel costs. In Scenario 3, solutions average an overall cost of 530.68, with travel costs averaging 243.20, compared to Scenario 1 where the average overall cost is 624.47 and travel costs average 357.80.
- The use of CLs also indicates potential reductions in the number of used EVs (i.e., the average number of used vehicles is 2.67 in Scenario 1 compared to 1.5 in Scenario 2 and 1.33 in Scenario 3), alleviating traffic congestion in urban areas.

6.2. *Impact of compensation rate*

To analyze how the difference in compensation rate affects the routing plans, another experiment is conducted by varying the compensation rate $\delta = \{0, 0.5, 1.5, 2, 5, 10, 15\}$. All customers are treated as Type 3 customers (i.e., served either through the home delivery option or the SP service) to ensure that the option selection depends on the compensation rate.

Table A6 lists the detailed results, where n_1 and n_2 represent the total number of customers served by Options 1 and 2, respectively. $f^{(1)}$ denotes the travel cost, and f is the overall cost. The results are plotted in Fig. 3a with various compensation rates. We observe the following.

- When δ increases, it becomes more beneficial for the last-mile delivery to shift customer assignments toward the AHD option. However, travel cost ($f^{(1)}$) and overall cost (f) show significant increases.

- At $\delta = 15$, it indicates a complete shift towards the AHD option; that is, EVRPTW-PR-CL results.
- Although employing a lower possible compensation rate results in a lower overall cost and travel costs, it may negatively influence customer incentives.

It is worth noting that decision makers may face a crucial trade-off between the minimization of operation costs and customer satisfaction. This highlights the need for a balance in the compensation rate.

6.3. Impact of coverage area

To explore the impact of the coverage area of CLs, we vary covering levels $\chi = \{1, 2, 3, 4\}$. Based on the covering radius equation in Section 5.1, it implies that when χ increases, it increases the number of customers served at the particular CL. At $\chi = 4$, each customer can be served by any CLs within the network. In addition, this experiment maintains the assumption that all customers are Type 3 customers.

Table A7 reports detailed results similar to Table A6. Figure 3b illustrates how different covering levels affect routing plan performance. We observe that a larger coverage area provides more flexibility for customer assignments, leading to a slight reduction in overall cost and a corresponding shift toward the SP service.

While higher covering levels offer operational advantages and potential cost savings, decision makers must balance covering radius to ensure customer satisfaction and delivery system practicality. Optimizing the last-mile delivery process requires finding this balance.

7. Conclusion

This research investigates an EVRPTW-PR-CL, adopting CLs equipped with PLs for facilitating SP services and CSs for recharging EV batteries. The objective function accounts for travel costs, compensation paid to customers, EV fixed costs, and fixed operational costs for CLs. We also consider an acceptance distance (i.e., the covering level) that customers are willing to pick up parcels. Our proposed problem considers three customer types, reflecting customer behaviors and responsibilities in promoting sustainability in last-mile delivery.

We formulate an MILP model to solve EVRPTW-PR-CL, which provides optimal solutions for instances with up to 10 customers and feasible solutions for instances with 15 customers within a two-hour time limit. We then develop a VNS algorithm with tailored neighborhood operators to solve large instances. Additionally, we test VNS on EVRPTW-PR benchmark instances, demonstrating that it outperforms state-of-the-art algorithms by providing new BKSs within a reasonable average computational time. Finally, we conduct sensitivity analysis to highlight the adoption of CLs in achieving cost savings. The compensation rate also significantly impacts routing decisions, and the covering level slightly adjusts routing plan flexibility.

While this paper focuses on compensation for customers using SP services, several future research directions can be explored. First, uncertainties in travel and charging times affected by external factors (e.g., traffic congestion and environmental temperature) can be incorporated to improve the robustness of the delivery process. Second, addressing the installation costs and capacities of CLs through LRPs (Drexl and Schneider, 2015) could contribute to more cost-effective solutions. Third, another extension involves offering multiple time windows for serving each customer at their associated locations to enhance the flexibility of routing plans (Dumez et al., 2021). Lastly, dynamic versions could be developed to handle real-time orders, enhancing flexibility and responsiveness in delivery systems.

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References

Bezerra, S.N., de Souza, S.R., Souza, M.J.F., 2023. A general VNS for the multi-depot open vehicle routing problem with time windows. *Optimization Letters* 17, 9, 2033–2063.

Bonomi, V., Mansini, R., Zanotti, R., 2022. Last mile delivery with parcel lockers: evaluating the environmental impact of eco-conscious consumer behavior. *IFAC-PapersOnLine* 55, 5, 72–77.

Boysen, N., Fedtke, S., Schwerdfeger, S., 2021. Last-mile delivery concepts: a survey from an operational research perspective. *OR Spectrum* 43, 1–58.

Brimberg, J., Salhi, S., Todosijević, R., Urošević, D., 2023. Variable neighborhood search: the power of change and simplicity. *Computers & Operations Research* 155, 106221.

Cantillo, V., Amaya, J., Serrano, I., Cantillo-García, V., Galván, J., 2022. Influencing factors of trucking companies willingness to shift to alternative fuel vehicles. *Transportation Research Part E: Logistics and Transportation Review* 163, 102753.

Çatay, B., Sadati, İ., 2023. An improved matheuristic for solving the electric vehicle routing problem with time windows and synchronized mobile charging/battery swapping. *Computers & Operations Research* 159, 106310.

Conrad, R.G., Figliozzi, M.A., 2011. The recharging vehicle routing problem. Proceedings of the 2011 Industrial Engineering Research Conference, Vol. 8, IISE, Norcross, GA.

Cortés-Murcia, D.L., Prodhon, C., Afsar, H.M., 2019. The electric vehicle routing problem with time windows, partial recharges and satellite customers. *Transportation Research Part E: Logistics and Transportation Review* 130, 184–206.

Desaulniers, G., Errico, F., Irnich, S., Schneider, M., 2016. Exact algorithms for electric vehicle-routing problems with time windows. *Operations Research* 64, 6, 1388–1405.

Deutsch, Y., Golany, B., 2018. A parcel locker network as a solution to the logistics last mile problem. *International Journal of Production Research* 56, 1–2, 251–261.

Dietl, M., Voigt, S., Kuhn, H., 2024. From rush to responsibility: evaluating incentives on online fashion customers' willingness to wait. *Transportation Research Part D: Transport and Environment* 133, 104280.

Drexl, M., Schneider, M., 2015. A survey of variants and extensions of the location-routing problem. *European Journal of Operational Research* 241, 2, 283–308.

Dumez, D., Lehuédé, F., Péton, O., 2021. A large neighborhood search approach to the vehicle routing problem with delivery options. *Transportation Research Part B: Methodological* 144, 103–132.

Enthoven, D.L., Jargalsaikhan, B., Roodbergen, K.J., Uit het Broek, M.A., Schrotenboer, A.H., 2020. The two-echelon vehicle routing problem with covering options: city logistics with cargo bikes and parcel lockers. *Computers & Operations Research* 118, 104919.

Erdelić, T., Carić, T., 2019. A survey on the electric vehicle routing problem: variants and solution approaches. *Journal of Advanced Transportation* 2019, https://doi.org/10.1155/2019/5075671.

Erdelić, T., Carić, T., 2022. Goods delivery with electric vehicles: electric vehicle routing optimization with time windows and partial or full recharge. *Energies* 15, 1, 285.

Froger, A., Jabali, O., Mendoza, J.E., Laporte, G., 2022. The electric vehicle routing problem with capacitated charging stations. *Transportation Science* 56, 2, 460–482.

Gevaers, R., Van de Voorde, E., Vanelslander, T., 2009. Characteristics of innovations in last-mile logistics—using best practices, case studies and making the link with green and sustainable logistics. European Transport Conference, Vol. 1. Association for European Transport, p. 21.

Goeke, D., Schneider, M., 2015. Routing a mixed fleet of electric and conventional vehicles. *European Journal of Operational Research* 245, 1, 81–99.

Grabenschweiger, J., Doerner, K.F., Hartl, R.F., Savelsbergh, M.W., 2021. The vehicle routing problem with heterogeneous locker boxes. *Central European Journal of Operations Research* 29, 113–142.

Grabenschweiger, J., Dörner, K.F., Hartl, R.F., 2022. The multi-period location routing problem with locker boxes. *Logistics Research* 15, 1, 1–25.

Hiermann, G., Hartl, R.F., Puchinger, J., Vidal, T., 2019. Routing a mix of conventional, plug-in hybrid, and electric vehicles. *European Journal of Operational Research* 272, 1, 235–248.

Hong, J., Lee, M., Cheong, T., Lee, H.C., 2019. Routing for an on-demand logistics service. *Transportation Research Part C: Emerging Technologies* 103, 328–351.

Jaller, M., Pahwa, A., 2020. Evaluating the environmental impacts of online shopping: a behavioral and transportation approach. *Transportation Research Part D: Transport and Environment* 80, 102223.

Kahr, M., 2022. Determining locations and layouts for parcel lockers to support supply chain viability at the last mile. *Omega* 113, 102721.

Keskin, M., Çatay, B., 2016. Partial recharge strategies for the electric vehicle routing problem with time windows. *Transportation Research Part C: Emerging Technologies* 65, 111–127.

Keskin, M., Çatay, B., 2018. A matheuristic method for the electric vehicle routing problem with time windows and fast chargers. *Computers & Operations research* 100, 172–188.

Keskin, M., Laporte, G., Çatay, B., 2019. Electric vehicle routing problem with time-dependent waiting times at recharging stations. *Computers & Operations Research* 107, 77–94.

Kin, M., 2022. 9 Mind-blowing worldwide e-commerce statistics. Technical Report. https://www.markinblog.com/ecommerce-statistics/.

Koyuncu, I., Yavuz, M., 2019. Duplicating nodes or arcs in green vehicle routing: a computational comparison of two formulations. *Transportation Research Part E: Logistics and Transportation Review* 122, 605–623.

Luo, R., Ji, S., Ji, Y., 2022. An active-learning Pareto evolutionary algorithm for parcel locker network design considering accessibility of customers. *Computers & Operations Research* 141, 105677.

Mancini, S., Gansterer, M., Triki, C., 2023. Locker box location planning under uncertainty in demand and capacity availability. *Omega* 120, 102910.

Mara, S.T.W., Kuo, R., Asih, A.M.S., 2021. Location-routing problem: a classification of recent research. *International Transactions in Operational Research* 28, 6, 2941–2983.

de Mello Bandeira, R.A., Goes, G.V., Gonçalves, D.N.S., Márcio de Almeida, D., de Oliveira, C.M., 2019. Electric vehicles in the last mile of urban freight transportation: a sustainability assessment of postal deliveries in Rio de Janeiro – Brazil. *Transportation Research Part D: Transport and Environment* 67, 491–502.

Mladenović, N., Hansen, P., 1997. Variable neighborhood search. *Computers & Operations Research* 24, 11, 1097–1100.

Montoya, A., Guéret, C., Mendoza, J.E., Villegas, J.G., 2017. The electric vehicle routing problem with nonlinear charging function. *Transportation Research Part B: Methodological* 103, 87–110.

Orenstein, I., Raviv, T., Sadan, E., 2019. Flexible parcel delivery to automated parcel lockers: models, solution methods and analysis. *EURO Journal on Transportation and Logistics* 8, 5, 683–711.

Ozbaygin, G., Karasan, O.E., Savelsbergh, M., Yaman, H., 2017. A branch-and-price algorithm for the vehicle routing problem with roaming delivery locations. *Transportation Research Part B: Methodological* 100, 115–137.

Ozbaygin, G., Savelsbergh, M., 2019. An iterative re-optimization framework for the dynamic vehicle routing problem with roaming delivery locations. *Transportation Research Part B: Methodological* 128, 207–235.

Ozyavas, P., Buijs, P., Ursavas, E., Teunter, R., 2024. Designing a sustainable delivery network with parcel locker systems as collection and transfer points. *Omega* 131, 103199.

Rastani, S., Çatay, B., 2023. A large neighborhood search-based matheuristic for the load-dependent electric vehicle routing problem with time windows. *Annals of Operations Research* 324, 761–793.

Reyes, D., Savelsbergh, M., Toriello, A., 2017. Vehicle routing with roaming delivery locations. *Transportation Research Part C: Emerging Technologies* 80, 71–91.

Ropke, S., Pisinger, D., 2006. An adaptive large neighborhood search heuristic for the pickup and delivery problem with time windows. *Transportation science* 40, 4, 455–472.

Sadati, M.E.H., Akbari, V., Çatay, B., 2022. Electric vehicle routing problem with flexible deliveries. *International Journal of Production Research* 60, 13, 4268–4294.

Schiffer, M., Walther, G., 2018. An adaptive large neighborhood search for the location-routing problem with intra-route facilities. *Transportation Science* 52, 2, 331–352.

Schneider, M., Stenger, A., Goeke, D., 2014. The electric vehicle-routing problem with time windows and recharging stations. *Transportation science* 48, 4, 500–520.

Schwerdfeger, S., Boysen, N., 2022. Who moves the locker? A benchmark study of alternative mobile parcel locker concepts. *Transportation Research Part C: Emerging Technologies* 142, 103780.

Statista, 2022. Retail e-commerce sales worldwide from 2014 to 2026. Technical Report, https://www.researchgate.net/figure/Worldwide-retail-sales-of-e-commerce-from-2014-to-2026-Statista-2022-3.

Stenger, A., Schneider, M., Schwind, M., Vigo, D., 2012. Location routing for small package shippers with subcontracting options. *International Journal of Production Economics* 140, 2, 702–712.

Subramanian, A., Drummond, L.M., Bentes, C., Ochi, L.S., Farias, R., 2010. A parallel heuristic for the vehicle routing problem with simultaneous pickup and delivery. *Computers & Operations Research* 37, 11, 1899–1911.

Tilk, C., Olkis, K., Irnich, S., 2021. The last-mile vehicle routing problem with delivery options. *OR Spectrum* 43, 4, 877–904.

UNCTAD, 2017. UNCTAD framework for sustainable freight transport. Technical Report, https://unctad.org/publication/unctad-framework-sustainable-freight-transport-sft-framework.

Veenstra, M., Roodbergen, K.J., Coelho, L.C., Zhu, S.X., 2018. A simultaneous facility location and vehicle routing problem arising in health care logistics in the netherlands. *European Journal of Operational Research* 268, 2, 703–715.

Vukićević, M., Ratli, M., Rivenq, A., Zrikem, M., 2023. Covering delivery problem with electric vehicle and parcel lockers: variable neighborhood search approach. *Computers & Operations Research* 157, 106263.

Yavuz, M., Çapar, I., 2017. Alternative-fuel vehicle adoption in service fleets: impact evaluation through optimization modeling. *Transportation Science* 51, 2, 480–493.

Yu, V.F., Anh, P.T., Baldacci, R., 2023a. A robust optimization approach for the vehicle routing problem with cross-docking under demand uncertainty. *Transportation Research Part E: Logistics and Transportation Review* 173, 103106.

Yu, V.F., Anh, P.T., Chen, Y.W., 2023b. The electric vehicle routing problem with time windows, partial recharges, and parcel lockers. *Applied Sciences* 13, 16, 9190.

Zhou, L., Baldacci, R., Vigo, D., Wang, X., 2018. A multi-depot two-echelon vehicle routing problem with delivery options arising in the last mile distribution. *European Journal of Operational Research* 265, 2, 765–778.

Zhou, L., Li, C., Hu, C., Du, J., 2023. Parcel locker location problem with selectable volume sizes and heterogeneous customers in the last mile delivery. *Transportation Letters* 16, 1–15.

Zhou, L., Wang, X., Ni, L., Lin, Y., 2016. Location-routing problem with simultaneous home delivery and customer's pickup for city distribution of online shopping purchases. *Sustainability* 8, 8, 828.

Appendix: Detailed computational results

Table A1
Combinations of shaking and improvement strategies

Combination	ζ^{shake}	ζ^{vnd}	ζ^{ls}	$\bar{f}$	$\bar{t}(s)$
1	**sequential**	random	first	1043.62*	183.13
2	random	random	first	1044.50	158.87
3	adaptive	random	first	1043.84	146.47
4	sequential	sequential	first	1044.24	99.65
5	**sequential**	**random**	first	1043.62*	183.13
6	sequential	adaptive	first	1045.50	126.73
7	**sequential**	**random**	**first**	1043.62*	183.13
8	sequential	random	best	1044.49	121.44
9	sequential	random	hybrid	1044.28	173.04

*Indicates the best average solution.
Bold texts denote the tuned strategies.

Table A2
Parameter setting for the proposed VNS

	Value 1	Value 2	Value 3	Initial value	Tuned value
η^{shake}_{max}	1	5	10	–	5
$\bar{f}$	1043.95	1043.62	1043.96		1043.62
$\bar{t}(s)$	134.12	183.13	189.7		183.13
$\eta^{penalty}$	10	30	50	10	10
$\bar{f}$	1043.62	1044.52	1044.01		1043.62
$\bar{t}(s)$	183.13	185.57	185.2		183.13
$\gamma^{penalty}$	1.05	1.1	1.15	1.15	1.05
$\bar{f}$	1043.55	1043.61	1043.62		1043.55
$\bar{t}(s)$	151.27	150.41	183.13		151.27

Table A3
Detailed results for large-sized benchmark EVRPTW-PR instances

Instance	Best solution of 10 runs (f_b)							VNS				Gap (Δf[%])						
	KÇ	SW	HHPV	CPA	YAC	EC	ÇS	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}(m)$	KÇ	SW	HHPV	CPA	YAC	EC	ÇS
c101	1043.38	1043.38	1044.51	1043.38	1043.38	1043.38	1043.38	1043.38	1043.38	0.55	0.34	0.00	0.00	−0.11	0.00	0.00	0.00	0.00
c102	1032.49	1029.44	1033.8	1017.7	1029.48	1079.12	1029.44	1017.70	1023.31	0.63	1.20	−1.45	−1.15	−1.58	0.00	−1.16	−6.04	−1.15
c103	973.39	971.86	1001.13	971.19	989.53	971.88	974.71	971.19	977.30	0.07	2.17	−0.23	−0.07	−3.08	0.00	−1.89	−0.07	−0.36
c104	886.72	884.38	893.04	884.38	924.20	884.63	884.98	884.38	884.96	0.23	2.52	−0.26	0.00	−0.98	0.00	−4.50	−0.03	−0.07
c105	1037.78	1048.06	1052.95	1015.79	1033.95	1081.23	1027.8	1015.79	1018.15	0.04	0.49	−2.16	−3.18	−3.66	0.00	−1.79	−6.44	−1.18
c106	1024.18	1010.56	1043.5	1009.33	1028.04	1070.88	1021.43	1008.03	1008.41	0.97	0.83	−1.60	−0.25	−3.52	−0.13	−1.99	−6.23	−1.33
c107	1058.11	1010.91	1013.76	1046.5	1036.17	1064.94	1078.42	1047.36	1057.54	0.43	0.85	−1.03	3.48	3.21	0.08	1.07	−1.68	−2.97
c108	1033.5	1031.85	1000.56	1022.48	1025.48	1025.16	1036.67	1022.48	1026.83	0.74	1.24	−1.08	−0.92	2.14	0.00	−0.29	−0.26	−1.39
c109	946.84	940.38	946.84	940.38	940.38	945.49	940.38	940.38	947.33	0.00	1.57	−0.69	0.00	−0.69	0.00	0.00	−0.54	0.00
c201	629.95	629.95	658.11	629.95	629.95	629.95	629.95	629.95	629.95	0.00	0.87	0.00	0.00	−4.47	0.00	0.00	0.00	0.00
c202	629.95	629.95	645.39	629.95	629.95	629.95	629.95	629.95	629.95	0.00	2.20	0.00	0.00	−2.45	0.00	0.00	0.00	0.00
c203	629.95	629.95	643.45	629.95	629.95	629.95	629.95	629.95	629.95	0.00	3.40	0.00	0.00	−2.14	0.00	0.00	0.00	0.00
c204	629.95	628.91	636.43	628.91	628.91	628.91	629.95	628.91	628.91	0.00	3.79	−0.16	0.00	−1.20	0.00	0.00	0.00	−0.16
c205	629.95	629.95	638.17	629.95	629.95	629.95	629.95	629.95	629.95	0.00	1.28	0.00	0.00	−1.31	0.00	0.00	0.00	0.00
c206	629.95	629.95	635.38	629.95	629.95	629.95	629.95	629.95	629.95	0.00	1.69	0.00	0.00	−0.86	0.00	0.00	0.00	0.00
c207	629.95	629.95	632.8	629.95	629.95	629.95	629.95	629.95	629.95	0.00	2.13	0.00	0.00	−0.45	0.00	0.00	0.00	0.00
c208	629.95	629.95	638.17	629.95	629.95	629.95	629.95	629.95	629.95	0.20	1.80	0.00	0.00	−1.31	0.00	0.00	0.00	0.00
r101	1636.69	1615.5	1630.14	1606.98	1642.86	1648.68	1613	1609.83	1613.04	1.65	0.53	−1.67	−0.35	−1.26	0.18	−2.05	−2.41	−0.20
r102	1461.38	1429.8	1521.33	1461.23	1471.88	1456.53	1432.68	1465.74	1489.97	1.09	1.21	0.30	2.45	−3.79	0.31	−0.42	0.63	2.26
r103	1262.75	1244.15	1264.81	1212.37	1261.76	1223.24	1249.03	1214.23	1227.48	0.51	1.63	−4.00	−2.46	−4.17	0.15	−3.91	−0.74	−2.87
r104	1078.99	1056.87	1089.92	1051.41	1081.43	1055.36	1065.26	1054.87	1060.21	0.53	1.77	−2.29	−0.19	−3.32	0.33	−2.52	−0.05	−0.98
r105	1373.94	1347.8	1396.8	1362.31	1375.04	1347.8	1374.06	1351.37	1358.47	0.29	0.62	−1.67	0.26	−3.36	−0.81	−1.75	0.26	−1.68
r106	1310.46	1268.25	1281.09	1256.19	1294.95	1265.18	1279.41	1263.13	1266.78	0.64	1.22	−3.75	−0.41	−1.42	0.55	−2.52	−0.16	−1.29
r107	1118.91	1110.95	1127.71	1108.47	1127.02	1106.56	1114.98	1104.51	1111.54	0.79	1.65	−1.30	−0.58	−2.10	−0.36	−2.04	−0.19	−0.95
r108	1031.14	1020.52	1042.8	1020.52	1039.59	1030.4	1030.48	1019.58	1027.59	1.90	1.34	−1.13	−0.09	−2.28	−0.09	−1.96	−1.06	−1.07
r109	1193.76	1186.99	1265.82	1185.77	1184.68	1182.84	1188.12	1181.59	1204.09	0.56	1.07	−1.03	−0.46	−7.13	−0.35	−0.26	−0.11	−0.55
r110	1090.92	1070.99	1095	1071.92	1092.22	1070.49	1077.41	1070.01	1075.99	1.81	1.75	−1.95	−0.09	−2.34	−0.18	−2.08	−0.04	−0.69
r111	1084.13	1072.21	1147.23	1072.46	1102.33	1085.18	1074.74	1071.35	1090.73	0.24	1.76	−1.19	−0.08	−7.08	−0.10	−2.89	−1.29	−0.32
r112	1017.31	1001.79	1013.95	1001.79	1001.79	1001.79	1011.86	1001.79	1004.16	1.55	2.01	−1.55	0.00	−1.21	0.00	0.00	0.00	−1.01

Continued

Table A3
(Continued)

Instance	Best solution of 10 runs (f_b)							VNS				Gap (Δf[%])						
	KÇ	SW	HHPV	CPA	YAC	EC	ÇS	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}(m)$	KÇ	SW	HHPV	CPA	YAC	EC	ÇS
r201	1262.1	1255.81	1261.64	1255.81	1280.65	1258.4	1261.64	1278.16	1297.91	0.43	2.81	1.26	1.75	1.29	1.75	−0.19	1.55	1.29
r202	1052.32	1051.48	1051.46	1051.46	1058.61	1053.13	1051.46	1055.65	1060.24	0.44	3.13	0.32	0.40	0.40	0.40	−0.28	0.24	0.40
r203	895.54	895.96	900.6	895.54	904.09	897.2	895.54	895.54	899.51	0.57	4.47	0.00	−0.05	−0.57	0.00	−0.95	−0.19	0.00
r204	720.15	779.49	783.53	780.91	793.88	780.72	780.14	779.57	783.99	0.71	9.51	7.62	0.01	−0.51	−0.17	−1.84	−0.15	−0.07
r205	987.36	988.55	987.36	987.22	987.18	989.69	987.36	987.22	994.20	0.29	2.58	−0.01	−0.13	−0.01	0.00	0.00	−0.25	−0.01
r206	922.7	922.83	924.48	922.7	927.36	929.39	922.7	922.19	924.90	0.47	3.75	−0.06	−0.07	−0.25	−0.06	−0.56	−0.78	−0.06
r207	846.59	843.2	846.53	857.07	853.23	850.56	846.59	845.80	849.80	0.22	8.32	−0.09	0.31	−0.09	−1.33	−0.88	−0.56	−0.09
r208	736.12	736.12	736.64	738.84	740.12	743.32	736.12	736.12	737.75	0.92	9.22	0.00	0.00	−0.07	−0.37	−0.54	−0.98	0.00
r209	868.95	863.36	867.8	870.68	880.45	876.87	866.67	863.22	871.15	0.20	3.53	−0.66	−0.02	−0.53	−0.86	−2.00	−1.58	−0.40
r210	843.36	846.33	845.27	846.62	849.69	848.53	843.21	845.83	847.55	1.04	3.83	0.29	−0.06	0.07	−0.09	−0.46	−0.32	0.31
r211	862.56	827.29	857.1	826.88	832.60	831.66	862.56	825.25	833.87	2.21	8.19	−4.52	−0.25	−3.86	−0.20	−0.89	−0.78	−4.52
rc101	1743.9	1648.99	1725.73	1661.53	1727.07	1677.28	1703.59	1663.11	1699.80	0.49	0.66	−4.86	0.85	−3.77	0.10	−3.85	−0.85	−2.43
rc102	1555.5	1510.16	1540.26	1510.16	1557.11	1517.43	1525.32	1516.88	1524.34	0.76	1.08	−2.55	0.44	−1.54	0.44	−2.65	−0.04	−0.56
rc103	1329.58	1304.35	1388.72	1346.83	1358.88	1353.28	1316.75	1359.34	1369.67	0.65	1.47	2.19	4.05	−2.16	0.92	0.03	0.45	3.13
rc104	1202.93	1175.06	1181.26	1175.83	1175.06	1179.2	1185.58	1175.06	1182.71	0.74	1.95	−2.37	0.00	−0.53	−0.07	0.00	−0.35	−0.90
rc105	1458.49	1450.82	1463.49	1446.3	1494.69	1470.81	1451.28	1441.69	1452.43	0.45	0.58	−1.17	−0.63	−1.51	−0.32	−3.68	−2.02	−0.67
rc106	1417.4	1385.96	1397.55	1383.14	1438.59	1397.63	1400.12	1386.36	1392.55	0.35	0.69	−2.24	0.03	−0.81	0.23	−3.77	−0.81	−0.99
rc107	1261.03	1250.3	1255.03	1244.83	1275.38	1255.91	1257.83	1240.21	1244.58	0.25	1.37	−1.68	−0.81	−1.19	−0.37	−2.84	−1.27	−1.42
rc108	1184.06	1154.14	1165.6	1159.9	1206.96	1161.86	1167.32	1154.14	1157.05	0.68	1.46	−2.59	0.00	−0.99	−0.50	−4.58	−0.67	−1.14
rc201	1446.84	1445.17	1446.03	1443.07	1448.78	1446.6	1446.42	1433.57	1443.31	0.98	1.61	−0.93	−0.81	−0.87	−0.66	−1.06	−0.91	−0.90
rc202	1416.96	1408.08	1434.18	1403.32	1431.88	1413.9	1416.96	1416.98	1430.91	0.86	3.80	0.00	0.63	−1.21	0.96	−1.05	0.22	0.00
rc203	1069.27	1060.32	1061.12	1068.28	1069.12	1062	1065.67	1054.91	1063.95	0.16	4.53	−1.36	−0.51	−0.59	−1.27	−1.35	−0.67	−1.02
rc204	886.23	884.75	887.1	884.97	904.66	889.39	884.41	884.21	885.65	1.85	4.93	−0.23	−0.06	−0.33	−0.09	−2.31	−0.59	−0.02
rc205	1262.22	1259.69	1289.08	1249.56	1263.53	1264.52	1259.67	1257.98	1281.29	1.10	3.61	−0.34	−0.14	−2.47	0.67	−0.44	−0.52	−0.13
rc206	1206.09	1189.11	1200.74	1187.4	1194.28	1207.98	1201.04	1185.09	1198.15	1.64	2.47	−1.77	−0.34	−1.32	−0.19	−0.78	−1.93	−1.35
rc207	993.26	997.04	985.67	996.63	987.93	1002.83	991.65	978.30	994.39	0.48	3.84	−1.53	−1.92	−0.75	−1.87	−0.98	−2.51	−1.36
rc208	839.71	833.12	836.93	833.12	837.69	842.78	839.71	833.12	837.14	0.61	4.64	−0.79	0.00	−0.46	0.00	−0.55	−1.16	−0.79
Average										0.62	2.55	−0.86	−0.03	−1.55	−0.06	−1.27	−0.78	−0.57

Table A4
Detailed results for small-sized EVRPTW-PR-CL instances

	GUROBI		VNS-noOption (1)				VNS-noSPF (2)				VNS (3)				GAP (Δf[%])		
	f	t	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	GUROBI	(1)	(2)
C101C5	453.49	2.98	453.49	453.49	0.00	0.05	453.49	453.49	0.00	0.01	453.49	453.49	0.00	0.07	0.00	0.00	0.00
C103C5	331.74	0.20	331.74	331.74	0.00	0.04	331.74	331.74	0.00	0.01	331.74	331.74	0.00	0.04	0.00	0.00	0.00
C206C5	374.32	3.67	427.59	427.59	0.00	0.04	374.32	374.32	0.00	0.02	374.32	374.32	0.00	0.10	0.00	−14.23	0.00
C208C5	311.79	0.95	367.32	367.32	0.00	0.04	311.79	311.79	0.00	0.02	311.79	311.79	0.00	0.07	0.00	−17.81	0.00
R104C5	341.09	0.81	341.09	341.09	0.00	0.05	341.09	341.09	0.00	0.01	341.09	341.09	0.00	0.05	0.00	0.00	0.00
R105C5	311.75	0.38	311.75	311.75	0.00	0.11	311.75	311.75	0.00	0.01	311.75	311.75	0.00	0.06	0.00	0.00	0.00
R202C5	348.06	18.17	351.51	351.51	0.00	0.05	348.07	348.07	0.00	0.01	348.07	348.07	0.00	0.04	0.00	−0.99	0.00
R203C5	343.06	67.37	403.75	403.75	0.00	0.05	343.06	343.06	0.00	0.02	343.06	343.06	0.00	0.06	0.00	−17.69	0.00
RC105C5	487.68	62.98	487.68	487.68	0.00	0.05	487.68	487.68	0.00	0.02	487.68	487.68	0.00	0.12	0.00	0.00	0.00
RC108C5	381.33	1.97	386.46	386.46	0.00	0.06	381.33	381.33	0.00	0.02	381.33	381.33	0.00	0.10	0.00	−1.34	0.00
RC204C5	355.29	67.77	385.67	385.67	0.00	0.06	355.29	355.29	0.00	0.03	355.29	355.29	0.00	0.07	0.00	−8.55	0.00
RC208C5	342.63	2.09	387.94	387.94	0.00	0.04	342.63	342.63	0.00	0.02	342.63	342.63	0.00	0.05	0.00	−13.22	0.00
C101C10	628.42	7204	768.19	768.19	0.00	0.11	628.43	628.43	0.00	0.09	628.43	628.43	0.00	0.17	0.00	−22.24	0.00
C104C10	487.57	7202	646.39	646.39	0.00	0.11	487.57	487.57	0.00	0.07	487.57	506.02	3.78	0.13	0.00	−32.57	0.00
C202C10	476.37	523.31	552.58	552.58	0.00	0.12	476.37	476.37	0.00	0.14	476.37	476.37	0.00	0.18	0.00	−16.00	0.00
C205C10	471.76	3.03	501.86	501.86	0.00	0.09	471.76	474.43	0.57	0.06	471.76	471.76	0.00	0.12	0.00	−6.38	0.00
R102C10	574.51	425.28	578.26	578.26	0.00	0.09	575.86	575.86	0.00	0.06	574.51	575.73	0.21	0.20	0.00	−0.65	−0.24
R103C10	401.77	1705.25	497.70	497.70	0.00	0.12	401.77	401.77	0.00	0.07	401.77	401.77	0.00	0.12	0.00	−23.88	0.00
R201C10	366.64	53.27	445.21	445.21	0.00	0.09	366.65	366.65	0.00	0.11	366.65	366.65	0.00	0.16	0.00	−21.43	0.00
R203C10	411.2	7204	510.33	510.33	0.00	0.12	411.20	411.20	0.00	0.15	411.20	411.20	0.00	0.20	0.00	−24.11	0.00
RC102C10	694.79	6.41	701.95	701.95	0.00	0.12	694.79	694.79	0.00	0.06	694.79	694.79	0.00	0.12	0.00	−1.03	0.00
RC108C10	724.5	7200	748.05	748.05	0.00	0.13	709.42	709.42	0.00	0.10	709.42	709.42	0.00	0.16	−2.13	−5.45	0.00
RC201C10	509.58	36.25	568.70	568.70	0.00	0.09	509.58	509.58	0.00	0.08	509.58	509.58	0.00	0.12	0.00	−11.60	0.00

Continued

Table A4
(Continued)

	GUROBI		VNS-noOption (1)				VNS-noSPF (2)				VNS (3)				GAP (Δf[%])		
	f	t	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	GUROBI	(1)	(2)
RC205C10	547.16	8.20	580.48	580.48	0.00	0.09	547.16	547.16	0.00	0.06	547.16	547.16	0.00	0.13	0.00	−6.09	0.00
C103C15	687.15	7200	732.62	732.62	0.00	0.53	687.16	687.39	0.03	0.28	687.16	687.16	0.00	0.49	0.00	−6.62	0.00
C106C15	651.61	7200	651.61	651.90	0.04	0.43	651.61	652.32	0.11	0.16	651.61	652.03	0.06	0.42	0.00	0.00	0.00
C202C15	664.29	7200	684.49	689.18	0.69	0.26	644.32	644.32	0.00	0.19	644.32	644.32	0.00	0.32	−3.10	−6.23	0.00
C208C15	663.7	4968	684.27	684.27	0.00	0.31	663.70	663.70	0.00	0.27	663.70	663.70	0.00	0.38	0.00	−3.10	0.00
R102C15	680.87	7200	721.41	721.41	0.00	0.22	680.87	680.87	0.00	0.26	680.87	682.77	0.28	0.45	0.00	−5.95	0.00
R105C15	654.55	7200	654.55	654.55	0.00	0.19	654.55	654.55	0.00	0.15	654.55	654.55	0.00	0.40	0.00	0.00	0.00
R202C15	547.53	7200	666.90	666.90	0.00	0.23	547.53	561.02	2.46	0.28	547.53	559.37	2.16	0.45	0.00	−21.80	0.00
R209C15	578.5	7200	643.82	643.82	0.00	0.25	578.50	580.11	0.28	0.36	578.50	580.11	0.28	0.45	0.00	−11.29	0.00
RC103C15	778.32	7200	896.10	896.10	0.00	0.45	778.32	784.15	0.75	0.19	778.32	781.23	0.37	0.39	0.00	−15.13	0.00
RC108C15	726.49	7200	768.74	768.74	0.00	0.19	730.41	730.46	0.01	0.20	726.49	730.04	0.49	0.30	0.00	−5.82	−0.54
RC202C15	599.36	7200	718.17	718.17	0.00	0.23	595.48	595.48	0.00	0.24	595.48	595.48	0.00	0.37	−0.65	−20.60	0.00
RC204C15	555.61	7200	589.99	589.99	0.00	0.27	555.62	555.62	0.00	0.46	555.62	555.62	0.00	0.53	0.00	−6.19	0.00
Average	512.9	3221.34	559.68	559.81	0.02	0.15	511.97	512.65	0.12	0.12	511.82	512.99	0.21	0.21	−0.16	−9.67	−0.02

Table A5
Detailed results for large-sized EVRPTW-PR-CL instances

	GUROBI	VNS-noOption (1)				VNS-noSPF (2)				VNS (3)				GAP (Δf[%])		
	f	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	GUROBI	(1)	(2)
c101	2737.63	3039.05	3048.67	0.32	13.86	2670.51	2714.10	1.61	58.04	2631.38	2676.78	1.70	68.55	−4.04	−15.49	−1.49
c102	2739.05	3231.10	3233.39	0.07	18.16	2643.72	2690.18	1.73	66.43	2643.72	2674.80	1.16	129.95	−3.61	−22.22	0.00
c103	2685.55	3446.95	3457.98	0.32	19.93	2642.20	2681.17	1.45	79.58	2631.27	2671.71	1.51	485.17	−2.06	−31.00	−0.42
c104	2547.50	2905.83	2921.14	0.52	25.91	2527.01	2567.84	1.59	78.62	2507.44	2550.78	1.70	479.09	−1.60	−15.89	−0.78
c105	2609.52	3385.45	3395.83	0.31	14.89	2588.41	2647.10	2.22	62.54	2587.49	2635.46	1.82	77.03	−0.85	−30.84	−0.04
c106	2613.23	3522.25	3525.10	0.08	16.01	2584.74	2627.74	1.64	67.08	2571.09	2622.34	1.95	132.01	−1.64	−36.99	−0.53
c107	2663.36	3092.77	3104.66	0.38	13.47	2606.42	2640.18	1.28	68.42	2606.42	2625.46	0.73	148.58	−2.18	−18.66	0.00
c108	2631.34	3100.72	3105.34	0.15	19.49	2595.68	2628.08	1.23	66.58	2595.68	2610.02	0.55	133.88	−1.37	−19.46	0.00
c109	2680.81	3240.65	3246.46	0.18	17.33	2610.41	2660.84	1.90	66.88	2610.41	2653.27	1.62	374.57	−2.70	−24.14	0.00
c201	1712.29	1783.71	1783.82	0.01	31.26	1671.88	1692.92	1.24	100.61	1665.04	1690.88	1.53	104.13	−2.84	−7.13	−0.41
c202	1638.49	1864.16	1877.05	0.69	41.65	1577.80	1631.20	3.27	128.18	1577.80	1629.93	3.20	131.03	−3.85	−18.15	0.00
c203	1750.33	2069.16	2070.50	0.06	51.57	1702.67	1716.36	0.80	131.36	1702.67	1716.11	0.78	134.59	−2.80	−21.52	0.00
c204	1541.93	1784.59	1790.44	0.33	36.18	1521.43	1545.59	1.56	145.65	1519.27	1542.41	1.50	149.04	−1.49	−17.46	−0.14
c205	1694.41	2122.30	2148.46	1.22	25.06	1676.93	1693.95	1.00	104.27	1669.62	1683.88	0.85	107.08	−1.48	−27.11	−0.44
c206	1648.51	1824.27	1837.34	0.71	26.26	1651.14	1679.34	1.68	119.45	1633.93	1668.95	2.10	123.46	−0.89	−11.65	−1.05
c207	1682.1	1971.73	1971.76	0.00	34.39	1643.52	1689.61	2.73	109.86	1643.52	1680.45	2.20	112.28	−2.35	−19.97	0.00
c208	1648.77	1913.20	1951.25	1.95	24.72	1637.72	1680.63	2.55	106.82	1637.72	1671.11	2.00	109.64	−0.67	−16.82	0.00
r101	2738.81	2993.42	3056.94	2.08	9.66	2723.05	2791.80	2.46	52.91	2677.18	2767.89	3.28	61.95	−2.30	−11.81	−1.71
r102	2554.63	2771.54	2784.14	0.45	13.34	2529.99	2583.60	2.08	55.38	2529.99	2557.18	1.06	61.83	−0.97	−9.55	0.00
r103	2565.85	2747.79	2811.43	2.26	19.97	2364.13	2443.48	3.25	63.03	2358.34	2427.99	2.87	70.45	−8.80	−16.51	−0.25
r104	2250.52	2504.37	2508.04	0.15	17.33	2204.46	2261.75	2.53	70.85	2204.46	2252.86	2.15	83.66	−2.09	−13.60	0.00
r105	2482.26	2738.41	2749.31	0.40	11.20	2398.36	2457.42	2.40	55.09	2387.95	2445.39	2.35	67.95	−3.95	−14.68	−0.44
r106	2440.22	2855.02	2862.91	0.28	13.76	2359.87	2417.37	2.38	59.34	2344.66	2396.79	2.17	68.74	−4.08	−21.77	−0.65
r107	2295.89	2786.63	2787.99	0.05	17.13	2271.81	2328.73	2.44	67.62	2268.95	2323.70	2.36	83.85	−1.19	−22.82	−0.13
r108	2367.35	2858.90	2866.79	0.28	19.29	2223.41	2254.59	1.38	71.39	2211.08	2248.52	1.67	87.72	−7.07	−29.30	−0.56

Continued

Table A5
(Continued)

	GUROBI	VNS-noOption (1)				VNS-noSPF (2)				VNS (3)				GAP (Δf[%])		
	f	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	GUROBI	(1)	(2)
r109	2538.90	2718.45	2724.67	0.23	13.03	2361.69	2433.71	2.96	64.41	2356.56	2426.56	2.88	73.15	−7.74	−15.36	−0.22
r110	2443.82	2649.89	2659.70	0.37	15.64	2309.54	2342.38	1.40	69.68	2284.65	2325.88	1.77	82.21	−6.97	−15.99	−1.09
r111	2506.65	2669.93	2679.74	0.37	21.89	2287.14	2332.09	1.93	72.49	2262.90	2309.98	2.04	83.00	−10.77	−17.99	−1.07
r112	2551.90	2661.47	2669.29	0.29	19.58	2276.85	2307.98	1.35	80.32	2261.68	2285.65	1.05	93.10	−12.83	−17.68	−0.67
r201	1917.96	2251.07	2257.75	0.30	23.96	1795.67	1845.45	2.70	139.40	1795.67	1844.21	2.63	146.69	−6.81	−25.36	0.00
r202	1782.08	1846.89	1883.14	1.92	35.38	1670.59	1708.11	2.20	203.39	1670.59	1708.11	2.20	209.57	−6.67	−10.55	0.00
r203	1705.94	2029.09	2035.96	0.34	44.52	1610.45	1662.39	3.12	191.17	1610.45	1662.28	3.12	188.59	−5.93	−26.00	0.00
r204	1484.11	2021.13	2022.05	0.05	47.11	1303.37	1341.37	2.83	302.80	1303.37	1339.61	2.71	298.35	−13.87	−55.07	0.00
r205	1783.72	1976.73	2019.57	2.12	26.78	1575.74	1630.44	3.36	192.13	1575.74	1630.43	3.35	189.98	−13.20	−25.45	0.00
r206	1756.40	1854.46	1875.13	1.10	47.34	1513.72	1539.62	1.68	221.02	1513.67	1539.20	1.66	216.82	−16.04	−22.51	0.00
r207	1548.94	1715.00	1724.46	0.55	50.92	1457.87	1466.68	0.60	312.66	1457.87	1466.68	0.60	306.08	−6.25	−17.64	0.00
r208	1395.85	1717.97	1720.88	0.17	53.44	1371.20	1411.87	2.88	295.06	1371.20	1411.81	2.88	290.71	−1.80	−25.29	0.00
r209	1684.27	1799.16	1842.99	2.38	37.08	1519.72	1542.28	1.46	219.78	1519.72	1542.18	1.46	216.17	−10.83	−18.39	0.00
r210	1687.84	1880.24	1918.65	2.00	38.44	1482.34	1544.17	4.00	218.74	1482.34	1542.16	3.88	221.22	−13.86	−26.84	0.00
r211	1500.72	1849.42	1850.25	0.04	38.28	1374.59	1422.54	3.37	303.21	1374.59	1422.41	3.36	306.43	−9.18	−34.54	0.00
rc101	2929.52	3233.90	3311.68	2.35	10.53	2962.39	3015.94	1.78	50.40	2912.02	2954.62	1.44	53.90	−0.60	−11.05	−1.73
rc102	2967.26	3180.79	3286.97	3.23	13.27	2911.88	2952.29	1.37	56.98	2903.95	2923.53	0.67	62.69	−2.18	−9.53	−0.27
rc103	3004.22	3187.13	3198.65	0.36	15.10	2774.63	2836.56	2.18	64.41	2774.15	2805.74	1.13	88.81	−8.29	−14.89	−0.02
rc104	2861.79	3130.61	3160.20	0.94	19.98	2695.25	2739.93	1.63	73.54	2684.88	2729.10	1.62	89.19	−6.59	−16.60	−0.39
rc105	3055.72	3460.96	3488.89	0.80	14.23	2909.99	2981.77	2.41	55.31	2869.30	2931.06	2.11	63.79	−6.50	−20.62	−1.42
rc106	2909.19	3092.77	3106.04	0.43	14.71	2824.45	2867.89	1.51	55.07	2776.06	2851.36	2.64	63.71	−4.80	−11.41	−1.74
rc107	2841.14	3357.52	3409.31	1.52	14.67	2657.84	2706.90	1.81	65.54	2657.84	2702.69	1.66	73.41	−6.90	−26.33	0.00
rc108	2775.51	3260.95	3327.37	2.00	18.56	2632.14	2676.76	1.67	69.88	2628.20	2672.92	1.67	80.36	−5.60	−24.08	−0.15
rc201	2264.88	2543.95	2575.43	1.22	21.03	2195.57	2228.74	1.49	99.22	2195.57	2223.48	1.26	98.02	−3.16	−15.87	0.00

Continued

Table A5
(Continued)

	GUROBI	VNS-noOption (1)				VNS-noSPF (2)				VNS (3)				GAP (Δf[%])		
	f	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	f_b	f_a	$\Delta f_{(a-b)}$	$\bar{t}$	GUROBI	(1)	(2)
rc202	2008.68	2580.92	2595.60	0.57	33.40	1919.89	1987.32	3.39	146.14	1917.99	1983.74	3.31	144.27	−4.73	−34.56	−0.10
rc203	1972.98	2384.92	2400.56	0.65	47.79	1854.48	1928.10	3.82	173.97	1854.48	1919.38	3.38	171.78	−6.39	−28.60	0.00
rc204	1708.46	2079.74	2086.15	0.31	48.85	1595.29	1643.90	2.96	234.65	1595.29	1643.78	2.95	231.06	−7.09	−30.37	0.00
rc205	2050.56	2448.45	2455.71	0.30	30.80	1969.60	2034.54	3.19	155.02	1969.60	2032.65	3.10	153.50	−4.11	−24.31	0.00
rc206	2059.59	2496.73	2509.35	0.50	25.96	1903.42	1958.32	2.80	140.90	1903.42	1956.12	2.69	140.41	−8.20	−31.17	0.00
rc207	1853.70	2480.33	2487.87	0.30	37.10	1777.85	1807.08	1.62	166.57	1777.85	1801.05	1.29	165.01	−4.27	−39.51	0.00
rc208	1775.92	2328.62	2372.25	1.84	50.18	1601.49	1641.89	2.46	191.75	1601.49	1641.89	2.46	187.45	−10.89	−45.40	0.00
Average	2218.72	2561.49	2581.30	0.76	26.45	2120.07	2165.44	2.15	120.39	2112.15	2154.55	2.03	149.57	−5.36	−22.03	−0.32

Table A6
Effect of compensation rate (δ) in the results of EVRPTW-PR-CL

Instance	$\delta = 0$				$\delta = 0.5$				$\delta = 1$				$\delta = 1.5$			
	n_1	n_2	$f^{(1)}$	f	n_1	n_2	$f^{(1)}$	f	n_1	n_2	$f^{(1)}$	f	n_1	n_2	$f^{(1)}$	f
C103C15	6	9	210.13	460.13	8	7	219.88	527.00	10	5	228.14	577.13	13	2	313.15	585.30
C106C15	4	11	136.49	286.49	7	8	149.10	386.20	7	8	149.10	473.31	7	8	149.10	560.41
C202C15	2	13	180.80	380.80	7	8	195.55	473.57	9	6	235.44	546.74	15	0	383.62	583.62
C208C15	0	15	144.41	344.41	9	6	210.38	436.45	14	1	274.18	458.72	14	1	274.18	475.99
R102C15	0	15	130.89	380.89	2	13	160.17	460.59	2	13	160.17	561.00	8	7	207.63	646.82
R105C15	3	12	163.09	463.09	6	9	166.58	529.26	8	7	218.62	574.23	8	7	193.61	621.83
R202C15	0	15	93.31	293.31	7	8	173.80	405.73	8	7	178.44	468.87	9	6	214.23	528.35
R209C15	1	14	139.23	339.23	8	7	202.44	402.58	15	0	313.24	413.24	15	0	313.24	413.24
RC103C15	2	13	230.41	530.41	9	6	252.56	633.00	9	6	244.64	709.21	13	2	360.52	738.86
RC108C15	1	14	191.87	491.87	11	4	261.06	556.50	12	3	291.71	595.57	12	3	290.17	613.32
RC202C15	1	14	190.41	390.41	6	9	235.42	497.91	12	3	362.46	552.18	12	3	341.44	567.99
RC204C15	0	15	120.27	320.27	11	4	212.56	407.42	13	2	262.32	438.01	14	1	286.16	442.86
Average	1.67	13.33	160.94	390.11	7.58	7.42	203.29	476.35	9.92	5.08	243.20	530.68	11.67	3.33	277.25	564.88

Instance	$\delta = 2$				$\delta = 5$				$\delta = 10$				$\delta = 15$			
	n_1	n_2	$f^{(1)}$	f	n_1	n_2	$f^{(1)}$	f	n_1	n_2	$f^{(1)}$	f	n_1	n_2	$f^{(1)}$	f
C103C15	13	2	313.15	592.69	13	2	313.15	637.00	15	0	348.46	648.46	15	0	348.46	648.46
C106C15	15	0	275.13	575.13	15	0	275.13	575.13	15	0	275.13	575.13	15	0	275.13	575.13
C202C15	15	0	383.62	583.62	15	0	383.62	583.62	15	0	383.62	583.62	15	0	383.62	583.62
C208C15	14	1	274.18	493.26	15	0	300.55	500.55	15	0	300.55	500.55	15	0	300.55	500.55
R102C15	8	7	207.63	693.21	12	3	318.92	819.43	15	0	412.78	912.78	15	0	412.78	912.78
R105C15	8	7	201.55	663.42	14	1	331.60	703.97	14	1	331.60	726.33	15	0	336.15	736.15
R202C15	15	0	358.01	558.01	15	0	358.01	558.01	15	0	358.01	558.01	15	0	358.01	558.01
R209C15	15	0	313.24	413.24	15	0	313.24	413.24	15	0	313.24	413.24	15	0	313.24	413.24
RC103C15	14	1	368.67	747.51	14	1	368.67	790.78	15	0	397.67	797.67	15	0	397.67	797.67
RC108C15	12	3	290.17	637.70	15	0	370.25	670.25	15	0	370.25	670.25	15	0	370.25	670.25
RC202C15	12	3	341.44	576.84	15	0	394.39	594.39	15	0	394.39	594.39	15	0	394.39	594.39
RC204C15	14	1	286.16	445.10	14	1	286.16	458.52	14	1	286.16	480.88	15	0	382.22	482.22
Average	12.92	2.08	301.08	581.64	14.33	0.67	334.47	608.74	14.83	0.17	347.65	621.77	15.00	0.00	356.04	622.71

Table A7
Effect of covering level (χ) in the results of EVRPTW-PR-CL

Instance	$\chi = 1$				$\chi = 2$				$\chi = 3$				$\chi = 4$			
	n_1	n_2	$f^{(1)}$	f	n_1	n_2	$f^{(1)}$	f	n_1	n_2	$f^{(1)}$	f	n_1	n_2	$f^{(1)}$	f
C103C15	10	5	249.82	593.00	10	5	228.14	577.13	10	5	228.14	577.13	10	5	228.14	577.13
C106C15	7	8	149.10	473.31	7	8	149.10	473.31	7	8	149.10	473.31	7	8	149.10	473.31
C202C15	9	6	235.44	546.74	9	6	235.44	546.74	9	6	235.44	546.74	9	6	235.44	546.74
C208C15	14	1	274.18	458.72	14	1	274.18	458.72	14	1	274.18	458.72	14	1	274.18	458.72
R102C15	8	7	236.54	616.18	2	13	160.17	561.00	5	10	101.55	490.42	5	10	101.55	490.42
R105C15	9	6	252.21	578.15	8	7	218.62	574.23	8	7	218.62	574.23	8	7	190.93	564.29
R202C15	9	6	224.41	503.94	8	7	178.44	468.87	8	7	178.44	468.87	8	7	178.44	468.87
R209C15	15	0	313.24	413.24	15	0	313.24	413.24	15	0	313.24	413.24	15	0	313.24	413.24
RC103C15	9	6	296.38	687.18	8	7	277.10	695.07	10	5	217.49	665.70	10	5	217.49	665.70
RC108C15	12	3	291.71	595.57	12	3	291.71	595.57	12	3	231.84	586.74	12	3	231.84	586.74
RC202C15	12	3	341.44	559.14	12	3	341.44	559.14	8	7	212.06	538.22	8	7	212.06	538.22
RC204C15	13	2	262.32	438.01	13	2	262.32	438.01	13	2	262.32	438.01	13	2	262.32	438.01
Average	10.58	4.42	260.56	538.60	9.83	5.17	244.16	530.09	9.92	5.08	218.54	519.28	9.92	5.08	216.23	518.45